

Lexicographic Orders of Well-Orderings of the Surreal Numbers

Set approximations, the singular birthday transition, set-length words,
class orders, and higher-order structure

Research report merged from four independent manuscripts
prepared for Vladimir Reshetnikov (sources 08, 09, 11 and 12 of batch 80)

Sources dated 2 October 2026 (source 11: 3 October 2026, UTC); merged 2 October 2026

Abstract

The phrase *all well-orders of the surreal numbers* has several inequivalent meanings. Well-orders of a fixed birthday fragment form a set; well-orders of arbitrary set-sized subsets form an ordinary class; well-orders of the whole surreal line are themselves classes, and their totality belongs to a higher logical level. We develop these settings and distinguish set-like from unrestricted class well-orders.

For an infinite set-sized alphabet X , the existing ProveIt theory gives $|\mathcal{W}(X)| = 2^{|X|}$ and the exact ordinal embedding spectrum $\beta < (|X|)^+$. We apply it to surreal birthday fragments. For arbitrary set-length surreal words, we give an explicit sign-sequence embedding into **No**, with birthday bounds, and identify the one-sided cuts created by limit-length words.

For the whole surreal line, we define lexicographic comparison directly on labeled class well-orders, without assuming ordinal enumerations or a class of class order types. Every such well-order has a canonical initial segment of type Ord; its remaining tail determines a convex fiber. In the set-like fragment, bounded-support permutations of one global enumeration form an ordinary class isomorphic to **No**, and interpolate every uniformly set-indexed strict cut of the entire fragment. The fragment contains every class linear order, admits binary class-word coding, and cannot have an exhaustive uniform class of set-codes. The unrestricted order has adjacent pairs, all arising from finite residual alphabets. An omitted-range construction exhibits a proper-class cut filled by a non-set-like well-order. In a full inaccessible-universe interpretation, the unrestricted order does not embed into its set-like fragment.

The proofs use ZF, ZFC, or GBC, as specified, with a further uniform nonembedding theorem in KM + GC. We give universe-relative and proof-assistant formulations, and propose further research on the higher-order structure.

[merge] This report is being merged from four manuscripts written independently on 2 October 2026 on the same question and delivered together in batch 80. This edition contains source 11 (the base, whose abstract is printed above) and source 12; sources 09 and 08 follow in the second write. Each shared theorem is printed once; source 12's versions of source 11's results are recorded in numbered notes, and its genuinely different arguments are kept as marked routes. Source 12 adds an exact dichotomy at birthday cutoffs: for an infinite cardinal κ and $\mu = 2^{<\kappa}$, the minimal-length enumeration order $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is equimorphic with the binary cube $\mathbb{B}_{\kappa \cdot \kappa}$ (ordinal product) exactly when κ is a singular strong limit, and with $\mathbb{B}_\mu$ otherwise, with $\kappa = \beth_\omega$ as an unconditional example. It also adds a second direct comparison of all class well-orders, by their largest common labelled prefix, and an exact external identification of both global orders in a full-class model at an inaccessible cardinal. Source 11's choiceless proof answers source 12's question on the weak-choice strength, and source 12's dichotomy answers in part a question of the repository's report on the reals. The set-sized general-alphabet layer re-proves that report and is printed with pointers to it, never as new.

Status. This is a mathematical research report with proofs, not a claim of machine verification or bibliographic priority. Existing general set-alphabet results are credited to the repository. The further constructions are proved here and should be evaluated by their arguments. The repository audit was targeted; no Lean build or remote change was made.

[merge] **Status of the merged report.** AI-assisted and unrefereed; not formalized. Source 12 calls itself AI-assisted on its title page; source 11 does not use the word (author lines: 11 “Research report”, 12 “Research report prepared for Vladimir Reshetnikov”). Neither claims historical priority, peer review or machine verification. Their finite checks verify finite instances only. No statement of this report is formalized; the formal status of the inputs it uses is in [section 1.5](#).

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1 About this report

1.1 Four manuscripts, one report

[merge] All four manuscripts of this report arrived in commit 4e270aa46 and were placed in ccc046989 (batch 80, cluster K3). This edition merges source 11 (the base; pin ae7c1e6aa, 2 October 2026, 18:03:53 PDT = 3 October 2026, 01:03:53 UTC, which is the date its title page gives; 2,807 lines, 35-page PDF, archive surreal_well_orders (1).zip; labels swo:, files 11-raw-orders-*) and source 12 (pin 63bb8b1d9, 18:01:32 PDT; 1,689 lines, 24-page PDF, archive surreal_well_orders.zip; labels swo:st:, files 12-singular-*). Sources 09 and 08 (labels swo:ec: and swo:sk:, files 09-core-* and 08-skeleton-*) are merged in the second write. The two archives sharing the name surreal_well_orders are not editions of one another: word 8-gram overlap 0.7%, no shared member file, different pins and result sets.

Source 11 is the base. Its global-choice equivalence is proved over GB without set choice (theorem 13.1); it covers set alphabets, set-length words, and set-like and unrestricted class well-orders; and it attributes the general set-alphabet theorems to the repository’s report on the reals, SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/ (cited below as the reals report, labels lwo:).

1.2 How the texts are printed

[merge] Source 11’s text is printed in full and in its own order. Source 12’s introduction and foundational section are printed whole; a result that a printed result already proves by the same argument is recorded in a numbered *Note* with the source’s statement, number and differences; a genuinely different proof is printed in full as a marked route; a result proved only by source 12 is printed in full where it belongs. Every label survives with its prefix; unlabelled numbered statements carry swo:sub:nnumber. Section D maps every numbered statement to its place here. Text written for the merge is marked [merge].

1.3 One notation for the sources

[merge] The sources’ symbols collide. The following convention is fixed before any of them is used; every renaming is listed.

Here	Meaning, and the sources’ symbols
GB, GBC	GB has a ZF set part, elementary class comprehension and class replacement, and <i>no</i> choice principle (found:sub:gbconvention); GBC = GB + GC. Source 12 analyses choice “over GB plus set-level AC” (theorem 13.2), written GB + AC in the merge text; source 11 works over GB itself.
bd(x)	Birthday. Source 11’s $\ell(x)$ and source 12’s $b(s)$ are renamed.
$\mathbf{No}_{<\kappa}$	$\{x \in \mathbf{No} : \text{bd}(x) < \kappa\}$. Renamed: source 11’s X_κ , source 12’s S_κ (the reals report’s $S_{<\kappa}$, not $S_{<\kappa+}$).
$\ell_2(L)$	Binary coding length (lwo:def:binary-length); source 12 prints the same symbol and calls it the “binary coding rank”. $\text{rank}(x)$ is the von Neumann rank (source 11).
$\mathcal{W}(X)$, $\mathcal{W}_\alpha(X)$	Exhaustive enumerations of a set X , and those of length α ; source 11’s $\mathcal{P}_\lambda(X)$ is renamed $\mathcal{W}_\lambda(X)$.
$\mathfrak{W}_{\text{sl}}(\mathbf{No})$, $\mathfrak{W}_{\text{all}}(\mathbf{No})$	Set-like and all class well-orders of $\mathbf{No}$, predicates on classes; source 12’s $\text{SL}(\mathbf{No}) = \mathcal{E}(\mathbf{No})$ and $\mathcal{W}_{\text{cl}}(\mathbf{No})$.
$\mathbb{B}_\gamma$, $\mathbb{B}_{\text{Ord}}$	Binary cubes; class binary words on Ord (source 12’s $\mathfrak{B}$).

Here	Meaning, and the sources' symbols
$\simeq_e$	Equimorphism; source 11's $\approx_{\text{emb}}$.
e_0, B	The baseline enumeration of sources 11 and 12.

1.4 What the report claims, and what it does not

[merge] The report claims the written proofs printed below, each in the foundation its statement names. None is claimed to be historically first. The set-sized general-alphabet layer re-proves the reals report and is printed with pointers to it (section 6.1); universality and homogeneity of **No** are classical (Ehrlich [15]; Hamkins [20]).

- *Source 11* (its Status paragraph, its delivery README's "Scope and verification" and its audit): no machine verification, no Lean build, no new Lean file; no bibliographic priority; research questions are proposed directions; the audit was targeted; the finite checks do not verify transfinite proofs, cardinal arithmetic or class comprehension. Its README states: "The mathematical proofs underwent a separate proof review." That is the delivery's own statement; no such review is recorded in the repository, and this report does not rely on it.
- *Source 12* (title page, its Section 1.2, and 12-singular-RESEARCH_STATUS.md, "What is not claimed"): AI-assisted, unrefereed; priority not claimed; no catalogued open problem claimed settled; no new Lean or Rocq verification; the Python checks are not substitutes for the proofs; the bounded theorem is not a construction of a full class model at a singular cardinal; the collection of class well-orders is not treated as a set or class at the same level; no universal semantic decoder is assumed; its research questions are proposals; its source search was not sufficient to establish originality.
- *Review status.* The repository's Hilbert's-tenth research tree holds a written-proof review of the four archives of this cluster (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md, commit c338541e3); it found their principal arguments sound under the stated foundations, with one scoped correction to source 08, which joins in the second write. That review is pinned to the delivered archives and does not cover the text written for this merge; it is not an entry of the surreal collection's review record, where this report's review is pending.

1.5 Formal status

[merge] No statement of this report is formalized, and no `swo:` label has a Lean mapping in the collection's formalization ledger (Algebra/SurrealNumbers/docs/FORMALIZATION.md). Placement in the surreal collection confers no formal status. Inputs the sources use are proved in Lean in universe-relative form: properness and bounded birthdays (`found:prop:proper`; `birthdays_bounded`, `small_bounded`, `not_small` in `SignSequence.lean`), small-cut filling (`small_cut_fillers`, `exists_separator_of_small_sets`) and the inconsistency of unrestricted cuts (`noUnrestrictedCuts`). The declarations cited by sources 11 and 12 exist at the current revision; none states a result of this report.

1.6 Where the merge had to choose

[merge]

- *Base and order.* Source 11; its section order is kept, and source 12's material follows the base result it duplicates or extends.

- *Strongest statement printed as the statement.* The choice equivalence is source 11’s (over GB; source 12 assumes set choice). The diagonal theorem is printed twice: source 11’s (theorem 18.19) and source 12’s stronger theorem 18.20. The birthday-cutoff statement is source 12’s dichotomy.
- *Different routes kept in full:* source 12’s set-sized spectra through binary coding length, its avoidance lemma, its proof of the choice equivalence over $\mathbf{GB} + \mathbf{AC}$, its largest common labelled prefix, its residual adjacency criterion, its binary-class concatenation, its prefix extension, and its birthday-bounded universality.
- *Corrections.* The renamings of section 1.3; one mistyped reference in source 11; source 12’s weak-choice question printed as answered (remark 22.15).

1.7 Where one source answers another

[merge] Dated notes, 2 October 2026.

- Source 12’s question on the weak-choice strength* (remark 22.15, and its remark 13.3): answered by source 11. Over GB without set choice, a class well-order of $\mathbf{No}$ exists exactly when global choice holds (theorem 13.1); see remark 13.4.
- The reals report’s question “Other ground orders”* (`lwo:sec:research`, which names “set-sized suborders of surreal numbers”): answered in part by theorem 8.3 for $X = \mathbf{No}_{<\kappa}$ and the minimal stratum: $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\mu$, except at singular strong-limit κ , where it is $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$ and not $\simeq_e \mathbb{B}_\kappa$. Open for longer strata and other alphabets (research question 22.11).
- Hamkins’s question whether universality of the surreal line is a weak global choice principle* [21]: not settled here, as source 12 says after its remark 22.15.

2 The objects being studied

There are two orders in this problem. Each object under consideration is itself a well-order of some collection of surreal numbers. We then compare these objects lexicographically using the *numerical* order on their surreal labels. This second order need not be a well-order. Indeed, the orders below contain abundant descending sequences.

Write $L \hookrightarrow M$ for an order embedding and $L \simeq_e M$ for embeddings in both directions. Equimorphism does not imply isomorphism.

Object	Elements	Logical size and interpretation
$\mathcal{W}(X)$	Ordinal-indexed bijections onto a fixed set $X \subseteq \mathbf{No}$	A set; ordinary ZFC order theory
$\mathcal{I}(\mathbf{No})$	Injective set-length ordinal words in $\mathbf{No}$	A class of sets; well-orders of arbitrary set-sized subsets
$\mathfrak{W}_{\text{sl}}(\mathbf{No})$	Set-like class well-orders of all $\mathbf{No}$, equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$	A higher-order totality; use class variables or uniform families
$\mathfrak{W}_{\text{all}}(\mathbf{No})$	All class well-orders of all $\mathbf{No}$, including non-set-like ones	A higher-order totality; direct labeled comparison is needed

The fraktur notation is deliberate. It does not declare a class with proper classes as members. Statements about $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{W}_{\text{all}}$ are formulas with class variables, or statements relative to a specified larger universe. An embedding of an ordinary class order into one of these totalities is a uniform family: one class relation encodes the object assigned to each set-valued index.

Principal answers. For a fixed infinite alphabet X , the exact ordinal ceiling is $(|X|)^+$. Passing to all set-length surreal words removes that ceiling: there is an explicit embedding of the entire class into **No**, and the singleton words embed **No** in the other direction. The two orders are nevertheless not isomorphic, because limit-length words have set-sized left cofinality.

Under global choice, the set-like global fragment contains a uniformly indexed copy of every class linear order. This includes every set ordinal, Ord, reverse Ord, Ord + Ord, and the whole numerical surreal order. Yet the entire fragment cannot be exhausted by one uniform class of set-codes. Class universality of a subfamily is compatible with a larger totality of class objects.

The unrestricted global order is different again. It has adjacent pairs and convex residual fibers. Every raw well-order begins with a canonical Ord-sequence; if that sequence omits labels, those labels form a later well-ordered tail. This decomposition permits a precise study of *all* well-orders without treating them all as set-like. It also supplies a nonembedding obstruction: the unrestricted order has too many disjoint two-point fibers to embed into the set-like order in the full containing-universe model. A related uniform definability obstruction is provable in KM + GC.

3 Relationship to ProveIt and the literature

We inspected the repository at the immutable revision

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The closest precursor is the report on lexicographic well-orderings of the reals [1], in the subdirectory `lexicographic-well-orderings-of-reals/` of the repository's ordinal-order reports. Its general alphabet theorems already establish the cardinality, universality, and ordinal cutoff for every infinite linearly ordered set X . Their application to birthday fragments is a transfer of existing results, not a new generic alphabet theorem.

That report also proves, with $\mathfrak{c} = 2^{\aleph_0}$,

$$\mathbf{No}_{<\theta} \hookrightarrow \mathcal{W}(\mathbb{R}) \quad (\theta < \mathfrak{c}^+), \quad \mathbf{No}_{<\mathfrak{c}^+} \not\hookrightarrow \mathcal{W}(\mathbb{R}).$$

The negative assertion follows from the all-plus chain of length $\mathfrak{c}^+$. It warns against taking an incoherent union of stagewise embeddings.

The repository's foundational report [3] supplies the relevant size discipline: individual surreals are set-coded, bounded fragments are sets, arbitrary class cuts need not be fillable, and quantifying over class functions does not create a class of those functions. Its local set-valued recursion arguments are directly useful. A nearby vector-space report uses the same distinction to avoid an unnecessary ETR hypothesis [5].

The sign-sequence presentation is standard [12, 14]. Absolute universality and homogeneity of the surreal order are part of the established theory discussed by Ehrlich [15]. We supply the short recursion arguments needed here.

The Kanovei–Shelah connection. There is a methodological connection, but the objects are different. Kanovei and Shelah's actual index consists of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary order on $\mathcal{P}(\mathbb{N})$ [16, Section 1]. The maps need not be injective and do not enumerate the entire real line. Their construction illustrates how a definable linear order on many codes can replace a definably selected well-order. The present problem shares that idea while introducing a different level of class-size complexity.

Contribution and attribution. The central developments here are explicit surreal-word compression, direct comparison of unrestricted class well-orders, their canonical initial-Ord decomposition, the bounded-support interpolation theorem, and the relation between omitted ranges and non-set-like tails. No claim about real-specific countability, separability, or gap spectra is transferred to surreal alphabets without proof.

4 The other source’s introduction

[merge] Source 12 opens with its own statement of the problem, of its relation to ProveIt, and of the Kanovei–Shelah connection. Its introduction is printed whole, followed by notes where the merge qualifies it. Every result it announces is printed, once, in the body.

4.1 Source 12: scope, predecessors, and the results at a glance

The phrase “well-orders of surreal numbers” hides several distinct objects. The usual numerical order of **No** is not a well-order. One may instead put an unrelated well-order on the entire surreal carrier, or consider numerical well-orders on subclasses, or work with an ordinary set of bounded-birthday surreals. This article treats all three viewpoints, but the principal object is the lexicographic comparison of *exhaustive well-orders of a fixed carrier*.

The supplied ProveIt repository contains a substantial predecessor, *Lexicographic Orders of the Well-Orderings of the Reals*, merging three reports and distinguishing the full enumeration order from its minimal-length slice [1]. The inspected README explicitly marks that report as unrefereed and not formalized. Its real-number spectra, binary coding, finite-tail geometry, and warnings about isomorphism versus equimorphism motivate the present work. The repository also contains a Lean development of simplest elements of convex sets of sign sequences [10]. These are useful foundations, not a formal verification of the new theorems below.

We pinned the audit to repository commit

63bb8b1d99359e7188abf0a33fb152e121858150.

The audit was targeted, not an exhaustive review of the repository. All essential arguments used here are written out, so no unreviewed research claim from the repository is required as a black box.

4.1.1 Three different orders

For an infinite set-sized linear order X of cardinality μ , define

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\}, \quad \mathcal{W}(X) = \bigcup_{\mu \leq \alpha < \mu^+} \mathcal{W}_\alpha(X).$$

They are ordered by their first unequal entry, using the given order of X , not the well-order being represented.

For the proper class **No**, let $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ denote, as a *predicate on classes*, the class bijections $E : \text{Ord} \rightarrow \mathbf{No}$. These represent exactly the set-like global well-orders. Let $\mathfrak{W}_{\text{all}}(\mathbf{No})$ denote the predicate of being an arbitrary class well-order relation on **No**. Neither displayed notation asserts the existence of a class having these class-sized objects as members.

The comparison on $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ uses a least ordinal of disagreement. The comparison on $\mathfrak{W}_{\text{all}}(\mathbf{No})$ uses a largest common *labelled* initial segment. The latter can be a proper class, and this is essential rather than a cosmetic extension.

4.1.2 What is proved, and what is not being claimed

The principal developments are the elementary common-prefix theorem ([theorem 14.7](#)), the global-choice equivalence ([theorem 13.2](#)), the non-enumerability theorem ([theorem 18.20](#)), and the singular transition ([theorem 8.3](#)). The full and minimal bounded spectra are proved in [sections 6.2](#) and [8](#). The difference between the dense set-like order and finite-tail geometry in the full order is proved in [sections 16.1](#) and [18.2](#).

These are mathematical claims with supplied proofs. We do not claim that they settle a previously catalogued open problem, that their exact formulations are historically new, or that they have been independently refereed. The proposed research questions at the end are precisely that: questions suggested by the results, not an assertion of a complete literature census.

4.1.3 The Kanovei–Shelah connection

Kanovei and Shelah order maps $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters, using a fixed lexicographic order on $\mathcal{P}(\mathbb{N})$ [[16](#), §1]. Their maps need not be bijections, and their index is not the literal collection of all well-orders of $\mathbb{R}$, still less all class well-orders of **No**. The useful common idea is to compare representations at their first difference instead of choosing one canonical representative. Their index construction remains set-sized. Moving to all global surreal well-orders creates an additional size level that their construction does not remove.

[merge] *On source 12's introduction.* Source 12 read the reals report's README and the Lean file `SignSequenceSimplicity.lean`, not the reals report's article; it therefore re-proved the set-sized patterns it needed. They are printed here as a marked route with pointers to the reals report ([sections 6.1](#) and [6.2](#)). The reals report's article has not changed since source 12's pin. Source 12's principal developments are printed as [theorem 14.7](#) (a second route to [theorem 14.2](#)), [theorem 13.2](#) (a second route to [theorem 13.1](#), over $\mathbf{GB} + \mathbf{AC}$), [theorem 18.20](#) (printed beside [theorem 18.19](#)) and [theorem 8.3](#) (new).

5 Surreal order facts and foundational conventions

5.1 Signs, birthdays, and set cuts

A surreal number is represented by a sign sequence

$$s : \alpha \longrightarrow \{-, +\}$$

of set-ordinal length α , its birthday $\text{bd}(s)$. Numerical comparison at the first disagreement uses

$$- < \text{termination} < +.$$

This differs from dictionary order with every proper prefix smaller than every extension. A sign sequence can be on either side of an extension, depending on its next sign.

For an ordinal θ , let

$$\mathbf{No}_{<\theta} = \bigcup_{\alpha < \theta} \{-, +\}^\alpha.$$

It is a set. The full **No** is proper-class-sized, because all-plus sequences embed all ordinals. Every set of surreals has bounded birthdays by Replacement.

Definition 5.1. A class linear order L has the *set-cut property*, called η_{Ord} here, if for all sets $A, B \subseteq L$ with $A < B$ there is x with $A < x < B$. Empty sides are allowed. For a cardinal κ , the set-sized notation η_κ requires the same property when $|A \cup B| < \kappa$.

The surreal cut construction supplies this property for $\mathbf{No}$ [14, Section 2]. A separator can be chosen canonically as the unique simplest separator. This statement concerns *sets* of options; applying it to the whole class against the empty right side would be inconsistent.

Lemma 5.2 (Deleting a set). *If $S \subseteq \mathbf{No}$ is a set, then $\mathbf{No} \setminus S$ has the set-cut property. Every nonempty open interval of $\mathbf{No}$ is a proper class.*

Proof. Let $A < B$ be sets in $\mathbf{No} \setminus S$. First choose x with $A < x < B$, and put

$$S_- = \{s \in S : s \leq x\}, \quad S_+ = \{s \in S : x < s\}.$$

Then $A \cup S_- < B \cup S_+$. A separator of this new set cut avoids both parts of S . The argument allows empty sides. If an interval (a, b) were a set S , applying the result to $\{a\} < \{b\}$ in $\mathbf{No} \setminus S$ would be impossible. $\square$

Choosing both separators by simplicity makes the deletion argument uniform in the set parameters without global choice.

[merge] *Third route to the first clause of lemma 5.2 (source 12, Lemma 6.2).* Source 12 avoids a forbidden set by using that open intervals of $\mathbf{No}$ are proper classes.

Lemma 5.3 (Avoiding a forbidden set). *If $A < B$ are sets of surreals and D is a set, then some separator z of A, B lies outside D .*

Proof. Choose u between A and B , and then v between $\{u\}$ and B . Every point of (u, v) separates A, B , and that interval is a proper class. It cannot be contained in D . With a fixed global well-order, choose the first acceptable point, making the operation uniform. $\square$

5.2 Which recursion principle is used?

Here GB has a ZF set part, elementary class comprehension, and class replacement; choice is stated separately. We use $\text{GBC} = \text{GB} + \text{GC}$, which includes set choice. Cardinality calculations and cardinal-valued cofinality statements throughout the article are read in ZFC, or in the set part of GBC. The explicitly definable sign code and the direct set-cut constructions retain their stated choice-free strength.

Ordinary recursion along Ord, with a *set* output at every stage and a rule using the set of earlier values and fixed class parameters, is available in GB. On each set ordinal perform the usual recursion with those parameters. Uniqueness makes the local solutions agree; elementary comprehension assembles their graph.

This is different from arbitrary elementary transfinite recursion, ETR, whose stages may themselves be classes. The latter supports iterated truth predicates and is stronger than GBC [22]. The prefix searches, partial permutations, and back-and-forth maps used here have set-sized stages.

[merge] The other sources fix their foundational conventions in their own sections, printed whole next. They agree with source 11's conventions, with one difference of base theory: when it analyses choice, source 12 works over $\text{GB} + \text{AC}$ (section 1.3).

5.3 Source 12: the surreal carrier and the foundational conventions

5.3.1 Sign sequences

We use the sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each individual sequence is a set. Its length $\text{bd}(s) = \text{lh}(s)$ is its birthday. At the first difference, signs and termination are compared by

$$- < \text{end} < +.$$

This is the usual numerical order, not the ordinary “shorter comes first” dictionary convention. Simplicity is the separate relation of being a proper sign prefix. The standard surreal construction and its simplicity structure are treated by Gonshor and Ehrlich [12, 15].

We shall use the following standard feature, also obtainable directly from the construction: whenever A, B are *sets* of surreals with $A < B$, there is a surreal z with $A < z < B$. There is a unique such separator of minimum birthday. Moreover, if every member of $A \cup B$ has birthday less than an ordinal γ , a separator can be constructed with birthday at most γ . This is interpolation, not Dedekind completeness: no least upper bound is being asserted.

Every set of surreals has a birthday bound, by Replacement. Every nonempty numerical interval contains a proper class of surreals. To see the latter without arithmetic, start with $a < b$ and recursively choose x_α between $\{a\} \cup \{x_\xi : \xi < \alpha\}$ and $\{b\}$. The previous values form a set at each stage. Thus one obtains distinct points for all ordinals α .

5.3.2 Well-orders and set-likeness

A class well-order on a class X is a strict linear class relation R for which every nonempty subclass has an R -least element. In GBC this agrees with the usual formulations using nonempty set subsets or absence of descending ω -sequences; the relationship is discussed in [27, §3]. We use strict relations throughout.

It is *set-like* if

$$\text{Pred}_R(x) = \{y \in X : yRx\}$$

is a set for every x . A proper-class set-like well-order is isomorphic to Ord : map x to the ordinal type of its set of predecessors. The range is an initial segment of Ord , and cannot be a set ordinal when X is proper. This construction uses only set-sized order types.

Not all class well-orders are set-like. The ordered sums $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, and $\text{Ord} + \text{Ord} + 1$ are examples. Their notation means ordered sums of class orders, not new von Neumann ordinals.

5.3.3 Foundational levels

Our set-sized results are theorems of ZFC. Our global results are in GBC, meaning Gödel–Bernays set theory with global choice. When analyzing the necessity of choice we work over GB plus ordinary, set-level AC. The full collection of global well-orders is handled by formulas with class variables, not by pretending that a proper class is an element of another GBC class.

Kelley–Morse theory allows stronger comprehension, but its classes still have sets, not proper classes, as members. Moving from GBC to KM does not by itself create a powerclass object of all subclasses of $\mathbf{No}$. One can instead use a larger set-theoretic universe, a set model with a specified collection of classes, or an explicitly higher-order theory. These are distinct interpretations and will be kept distinct.

5.3.4 Why an ordinal-length recursion here need not invoke ETR

A recursion on Ord that produces one *set* at each stage and uses the set of previous values is available by the ordinary recursion theorem with class parameters in GB. In contrast, ETR permits a genuinely class-valued recursive stage, represented by a class of pairs of a stage and an arbitrary

set. Even a truth hierarchy brings stronger principles into play [27, §2]. Our forth constructions, block lengths, and canonical enumeration of a class with set-sized initial segments are set-valued recursions. We do not invoke a universal satisfaction predicate or unrestricted class-valued recursion.

5.4 Universality and uniqueness of the saturated class order

Proposition 5.4. *In ZFC, every set linear order embeds into $\mathbf{No}$. In GBC, every class linear order embeds uniformly into $\mathbf{No}$. Any two proper-class η_{Ord} orders are isomorphic in GBC.*

Proof. Enumerate the source in a cardinal-length well-order, or in type Ord for a proper class, using global choice. At stage α , the earlier images form a set. Separate the images of earlier points below the new point from those above it, and choose a separator. This extends an order embedding. Each partial map is a set.

For uniqueness, globally enumerate both class orders and alternately insert the first unused point from each enumeration. The appropriate set cut in the other order is fillable. Global choice selects a separator when none is distinguished. The union of the partial isomorphisms is a surjective class order isomorphism. $\square$

The same construction, started from a specified set-sized partial isomorphism, gives set homogeneity. These are the order-theoretic universality and homogeneity features of $\mathbf{No}$ [15]. The assertion that every class order embeds is made with global choice; it is not used as an unproved converse to global choice.

[merge] *Source 12's version (its Lemmas 6.5 and 6.6).* It exposes the size of the forth recursion, with a birthday bound for set orders, and proves the class isomorphism $\mathbf{No} \setminus A \cong \mathbf{No}$ that source 11 uses in proposition 18.11.

Lemma 5.5 (Surreal universality, with the size of the recursion exposed). *In GBC, every class linear order admits a class order-embedding into $\mathbf{No}$. For a set order enumerated in type λ , one can arrange that the image of its ξ th enumerated element has birthday at most ξ , for each $\xi < \lambda$.*

Proof. Use a global-choice enumeration of the carrier, either by an ordinal or by Ord. When the next element is considered, its previously embedded predecessors and successors form two sets $A < B$. Send it to the simplest surreal separator. It is distinct from all previous images and preserves order. Every stage has set-sized history. Induction gives the birthday bound, because the earlier birthdays are less than the current stage. This is the standard forth argument behind surreal universality [15, 20]. $\square$

Lemma 5.6 (Deleting a set does not change the numerical class order). *In GBC, $\mathbf{No} \setminus A$ is order-isomorphic to $\mathbf{No}$ for every set A . Every nonempty open numerical interval is also order-isomorphic to $\mathbf{No}$.*

Proof. Each target has interpolation for arbitrary separated sets, by lemma 5.3, adding interval bounds when needed. Each is a proper class and has an Ord-enumeration by global choice. Carry out a back-and-forth between the two enumerations. At each ordinal stage, extend the partial isomorphism to the next unused domain point and then to the next unused target point. The previous partial map is a set, and the required cut can be filled while avoiding its set-sized range. At the end the class map is onto and order-preserving. $\square$

Part I

Sets of surreals

6 Well-order codes on a fixed set alphabet

Let X be an infinite linearly ordered set, and put $\lambda = |X|$. We work in ZFC throughout this section. A well-order on the whole carrier X has a unique increasing enumeration $f : \alpha \rightarrow X$, where increasing refers to that well-order, not to the native order of X . Its domain satisfies $|\alpha| = \lambda$, and hence $\alpha < \lambda^+$. Write $\mathcal{W}(X)$ for these bijective enumeration codes, and introduce two related sets:

$$\mathcal{I}(X) = \bigcup_{\alpha < \lambda^+} \{f : \alpha \hookrightarrow X\}, \quad \mathcal{W}_\lambda(X) = \{f : \lambda \rightarrow X : f \text{ is bijective}\}.$$

Thus $\mathcal{I}(X)$ codes well-orders of arbitrary subsets of X , including the empty subset, whereas $\mathcal{W}(X)$ always uses the whole carrier. All three sets inherit the lexicographic order defined earlier. A proper prefix is smaller than its extension in $\mathcal{I}(X)$. Two distinct members of $\mathcal{W}(X)$ cannot be proper prefixes of one another: the shorter enumeration would already have used every element of X .

The repository's real-well-ordering report already proves the exhaustive and cardinal-length statements for arbitrary infinite alphabets [1]. We record those results together with the injective-word variant, and include a common proof to keep the size distinctions explicit. Neither density nor completeness of the native order is needed.

Theorem 6.1 (Cardinality and exact ordinal spectrum). *Each of $\mathcal{W}_\lambda(X)$, $\mathcal{W}(X)$, and $\mathcal{I}(X)$ has cardinality 2^λ , contains a copy of every linear order of cardinality at most λ , and contains neither λ^+ nor $(\lambda^+)^*$. Consequently the ordinals embeddable in any of them are exactly the ordinals $\alpha < \lambda^+$.*

Proof. There are at most λ^+ permitted domains. For each domain there are at most $\lambda^\lambda = 2^\lambda$ functions, so $|\mathcal{I}(X)| \leq \lambda^+ \cdot 2^\lambda = 2^\lambda$. For the reverse inequality, partition X into pairs $\{a_\xi, b_\xi\}$, $\xi < \lambda$, with $a_\xi < b_\xi$. The lexicographic order on $\lambda \times \{0, 1\}$, taking ξ first, has order type $2 \cdot \lambda = \lambda$. For $\varepsilon \in 2^\lambda$, list the ξ -th pair as (a_ξ, b_ξ) if $\varepsilon(\xi) = 0$, and as (b_ξ, a_ξ) otherwise. The resulting map

$$2^\lambda \rightarrow \mathcal{W}_\lambda(X)$$

is an order embedding when 2^λ has lexicographic order. This proves all three cardinality assertions.

Let L be any linear order of cardinality at most λ , and fix an injection $\iota : L \hookrightarrow \lambda$. Associate to $t \in L$ the binary function c_t whose value at $\iota(s)$ is 1 precisely when $s < t$, and whose remaining coordinates are 0. If $t < u$, then $c_t \leq c_u$ coordinatewise, and the inequality is strict at $\iota(t)$. Their first difference therefore has the correct direction. The preceding pair construction embeds L into $\mathcal{W}_\lambda(X)$.

It remains to exclude longer monotone sequences. Suppose that $(f_\xi)_{\xi < \lambda^+}$ is strictly increasing or strictly decreasing in $\mathcal{I}(X)$. We recursively construct eventual common prefixes p_α of length α , for $\alpha < \lambda^+$. At a limit stage, take the union of the earlier prefixes. Regularity of λ^+ ensures that a tail of the original sequence extends that union. At a successor step, at most one member of this tail equals p_α ; discard it and all earlier indices. The values at coordinate α are now monotone. A monotone sequence indexed by a tail of λ^+ , taking at most λ values, is eventually constant: the first occurrence indices of all its values have a supremum below λ^+ . Append this eventual value to p_α . Every constructed prefix is injective because it occurs in members of the original sequence. The union of the prefixes through λ^+ would therefore be an injection $\lambda^+ \hookrightarrow X$, a contradiction. Finally, an ordinal at least λ^+ contains λ^+ , whereas every smaller ordinal has cardinality at most λ . $\square$

The distinction between the models also affects their elementary order topology. The order $\mathcal{W}_\lambda(X)$ is dense. Indeed, let $f <_{\text{lex}} g$, with first difference at ξ . If the tail of f after ξ has an ascent, interchanging the two entries produces a point strictly between f and g . If the tail of g has a descent, the corresponding interchange works in the other direction. If neither exists, the first remaining tail is strictly decreasing and the second strictly increasing. Their common set of entries, after removing the two entries at ξ , is infinite and would be both well-ordered and reverse-well-ordered. This is impossible.

By contrast, $\mathcal{W}(X)$ has adjacent pairs. Choose $a < b$ in X , enumerate $X \setminus \{a, b\}$ as a prefix p , and compare $p \frown \langle a, b \rangle$ with $p \frown \langle b, a \rangle$. Any intervening enumeration would have to share p , after which only these two orders remain. Thus a cardinal-length convention changes the resulting linear order even though it does not change its cardinality or ordinal embedding spectrum.

Proposition 6.2 (Exactly where jumps occur). *Suppose $f <_{\text{lex}} g$ in $\mathcal{W}(X)$, with common prefix p up to their first difference. Put $Y = X \setminus \text{ran}(p)$, $a = f(\text{dom}(p))$, and $b = g(\text{dom}(p))$, so $a < b$. The pair is adjacent precisely when Y is finite, a, b are consecutive in its native order, the tail of f after a lists $Y \setminus \{a\}$ decreasingly, and the tail of g after b lists $Y \setminus \{b\}$ increasingly.*

Proof. If an unused element lies strictly between a and b , start an enumeration with that element after p and complete it; this lies between f and g . Likewise, an ascent after a , or a descent after b , can be interchanged to create an intervening enumeration. Adjacency thus forces the two remaining tails to be decreasing and increasing, respectively. Their common native suborder $Y \setminus \{a, b\}$ is then both well-ordered and reverse-well-ordered. It must be finite: an infinite well-order contains its first ω elements, a subset with no greatest element. This proves necessity. Conversely, an intervening enumeration must share p . There is no possible next entry strictly between a and b . Among enumerations beginning with a , the displayed decreasing continuation is the largest; among those beginning with b , the increasing continuation is the smallest. Neither branch admits an intervening point. $\square$

This description also identifies the finite components generated by adjacency. Write the domain of an enumeration uniquely as $\theta = \delta + n$, with δ a limit ordinal and $n < \omega$. Its component consists of the $n!$ permutations of the final n entries, keeping the prefix of length δ fixed. In particular, words of limit length have no immediate neighbor. A jump cannot cross from one such component to another, because its common-prefix remainder would be infinite.

Endpoints can be classified just as concretely. The least element of $\mathcal{W}(X)$, if one exists, must always choose the least unused native element: otherwise interchanging that entry with a later smaller one decreases the enumeration. Hence a least element exists exactly when the native order on X is a well-order, and is its increasing enumeration. The greatest-element statement is the reverse assertion. For $\mathcal{W}_\lambda(X)$, the same argument shows that a least element exists exactly when the native order type is the initial ordinal λ , not merely an arbitrary ordinal of cardinality λ . For example, an alphabet of native type $\lambda + 1$ has a least full enumeration, but no least λ -length permutation.

6.1 Pointers to the reals report

[merge] The theorems of this section hold for every infinite linearly ordered set and are theorems of the repository’s reals report, which proves them in that generality. Source 11 says so (section 3); its repository audit (11-raw-orders-repository_audit.md) adds that their application to birthday fragments “should be attributed as applications of existing repository results, not marketed as new general theorems”. Source 12 re-proves them; they are printed with these pointers, and no novelty is claimed for any of them.

Statement	Here	Reals report
Prefix-freeness	before theorem 6.1 ; lemma 6.3	<code>lwo:lem:prefix-free</code>
Cardinality; pair swaps	theorem 6.1 ; lemma 6.6 and theorem 6.8	<code>lwo:prop:cardinality</code> , <code>lwo:lem:all-small-cubes</code>
Cut coding	theorem 6.1 ; lemma 6.5	<code>lwo:lem:cut-code</code> , <code>lwo:thm:ordinal-spectrum(a)</code>
No λ^+ -sequence	theorem 6.1 ; lemma 6.7	<code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code>
Strict cubes	lemma 6.4	<code>lwo:thm:cube-strict</code>
Coding length λ^+	theorem 6.8	<code>lwo:thm:full-length</code>
Extrema; adjacency; $n!$ blocks	paragraphs after proposition 6.2	<code>lwo:lem:extrema</code> , <code>lwo:thm:adjacency</code> , <code>lwo:thm:blocks</code>

The arguments are the reals report's: pair swaps, cut codes, prefix stabilization and cylinder fusion. Source 12's Section 9 is printed in full because the singular transition of [section 8](#) uses its lemmas.

6.2 Source 12's route: a general set-sized spectrum theorem

We temporarily forget the surreal structure. Let X be any infinite linearly ordered set, and write $|X| = \mu$, an initial cardinal. The spectra below depend only on the cardinality, not on density, separability, or field operations of X .

Lemma 6.3 (Prefix-freeness of exhaustive enumerations). *Two distinct exhaustive injective ordinal enumerations of X are never proper prefixes of one another. Hence first-difference lexicography is unambiguous on $\mathcal{W}(X)$.*

Proof. An exhaustive prefix has already used every element of X . An injective proper extension would have to use a new element, which is impossible. $\square$

For any ordinal γ , let $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$. For linear orders L, M , write $L \hookrightarrow M$ for an order-embedding and $L \simeq_e M$ for embeddings in both directions. Define

$$\ell_2(L) = \min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$$

when such a binary coding ordinal exists. This is an ordinal-valued rank, not a cardinal invariant.

Lemma 6.4 (Strict binary hierarchy). *For ordinals α, β ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

In particular $\ell_2(\mathbb{B}_\alpha) = \alpha$.

Proof. Zero-padding proves the forward construction when $\alpha \leq \beta$. It suffices to rule out an embedding $f : \mathbb{B}_{\beta+1} \rightarrow \mathbb{B}_\beta$.

Recursively construct an input prefix s_ξ of length ξ and an output prefix t_ξ of length ξ , so that every input extending s_ξ is sent to an output extending t_ξ . At a successor stage, split the input cylinder into its two next-bit halves and split the output cylinder likewise. The two input halves have strictly ordered images. They cannot both have images meeting both output halves: an image point in the upper output half from the lower input half would exceed an image point in the lower

output half from the upper input half. Thus one input half has image contained in a single output half. Choose those halves to continue.

At a limit take unions; the corresponding cylinders are nonempty and the containment persists. At stage β , two inputs remain, differing in their last bit, but their images have the same full β -long output. This contradicts injectivity. For $\alpha > \beta$, padding embeds $\mathbb{B}_{\beta+1}$ into $\mathbb{B}_\alpha$, giving the general impossibility. $\square$

Lemma 6.5 (Characteristic-cut coding). *Every linear order of cardinality at most μ embeds in $\mathbb{B}_\mu$.*

Proof. Enumerate its carrier by distinct coordinates d_ξ , padding unused coordinates if necessary. Send x to the bits $\mathbf{1}[d_\xi < x]$. For $x < y$, every differing coordinate changes from 0 to 1, and the coordinate representing x does change. Thus the least difference gives the correct order. $\square$

Lemma 6.6 (Pair orientation coding). $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$. *More generally, $\mathbb{B}_\gamma \hookrightarrow \mathcal{W}(X)$ for every $\gamma < \mu^+$.*

Proof. Partition X into μ unordered pairs, sort each pair by X 's order, and orient the pair according to its bit. The ordinal sum of μ two-element blocks has type $2 \cdot \mu = \mu$.

For a general $\gamma < \mu^+$, choose $|\gamma|$ disjoint pairs and retain a disjoint reserve of size μ . Such a choice is possible even when $|\gamma| = \mu$. List the pairs in type γ , then well-order all remaining labels in type μ . The resulting enumeration has length $2 \cdot \gamma + \mu < \mu^+$. Its comparison is decided by the first oriented pair at which the inputs differ. $\square$

Lemma 6.7 (No μ^+ -long monotone chain). *Neither μ^+ nor its reverse embeds into $\mathcal{W}(X)$.*

Proof. Suppose $(e_i)_{i < \mu^+}$ is strictly increasing. We recursively stabilize its entries on final segments of the index set. Assume that a final segment agrees on a prefix p of length $\delta < \mu^+$. No enumeration in a further infinite final segment can end there, by lemma 6.3. Their δ th entries form a nondecreasing μ^+ -sequence in a set of size μ .

Such a sequence is eventually constant: choose one first occurrence for each value in its range; their at most μ indices have supremum below μ^+ by regularity. Past that supremum no larger value can first appear. At a limit stage $\delta < \mu^+$, take the supremum of the earlier stabilization indices, still below μ^+ , and the union of the earlier prefixes.

This would produce an injective sequence of labels from X of length μ^+ : every set-sized earlier part is shared by a final segment of injective enumerations. That contradicts $|X| = \mu$. Replace nondecreasing by nonincreasing for the reverse order. $\square$

Theorem 6.8 (Cardinality and exact spectra). *For every infinite linearly ordered set X of cardinality μ ,*

$$\begin{aligned} |\mathcal{W}(X)| &= |\mathcal{W}_\mu(X)| = 2^\mu, \\ \beta \hookrightarrow \mathcal{W}(X) &\iff \beta < \mu^+, \\ \beta \hookrightarrow \mathcal{W}_\mu(X) &\iff \beta < \mu^+, \\ \mathbb{B}_\gamma \hookrightarrow \mathcal{W}(X) &\iff \gamma < \mu^+, \\ \ell_2(\mathcal{W}(X)) &= \mu^+. \end{aligned}$$

The ordinal statements also hold for reversed ordinals, and every linear order of cardinality at most μ embeds even into $\mathcal{W}_\mu(X)$.

Proof. Pair coding gives 2^μ elements, while all well-orders are relations on X , giving the upper bound $2^{|X \times X|} = 2^\mu$. The lower ordinal bounds and universality follow from [lemmas 6.5 and 6.6](#); [lemma 6.7](#) gives the upper bounds. The same characteristic coding applies to reversed ordinals.

For the cube spectrum use [lemma 6.6](#). If $\gamma \geq \mu^+$, $\mathbb{B}_\gamma$ contains an increasing chain of type μ^+ : use the sequences that are 1 before a moving cut and 0 afterward. Thus it cannot embed into $\mathcal{W}(X)$.

Embed X into $\mathbb{B}_\mu$ using [lemma 6.5](#). Encode an exhaustive enumeration of length $\alpha < \mu^+$ by concatenating its μ -long entry codes, then pad with zeroes to length μ^+ . The unpadded length $\mu \cdot \alpha$ has cardinality at most μ , so is less than μ^+ . Distinct enumerations disagree before either ends, and their encoded first disagreement preserves the order. This proves $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\mu^+}$.

If it embedded in $\mathbb{B}_\rho$ for $\rho < \mu^+$, then the copy of $\mathbb{B}_{\rho+1}$ supplied by [lemma 6.6](#) would contradict [lemma 6.4](#). Hence the rank is exactly μ^+ . $\square$

Remark 6.9 (What the rank does not say). Although $\ell_2(\mathcal{W}(X)) = \mu^+$, the cube $\mathbb{B}_{\mu^+}$ does *not* embed back into $\mathcal{W}(X)$. An order can first admit a binary encoding at an ordinal without containing the full cube of that ordinal. Nor is the absence of μ^+ and its reverse claimed sufficient for an arbitrary order to embed in $\mathcal{W}(X)$.

7 Birthday cutoffs and their set-sized orders

Use the sign presentation of surreal numbers: $x \in \mathbf{No}$ is a function from an ordinal $\text{bd}(x)$ to $\{-, +\}$. At the first difference, the comparison is $- < \text{termination} < +$. This presentation and the fundamental set-cut theorem are treated in Gonshor and in the modern account of Bagayoko and van der Hoeven [\[12, 14\]](#). For an infinite cardinal κ , define

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}.$$

This is a set. Counting sign sequences gives

$$|\mathbf{No}_{<\kappa}| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}.$$

Here $2^{<\kappa}$ denotes the supremum of 2^μ over cardinals $\mu < \kappa$; it is at least κ .

Corollary 7.1. *Put $\lambda = 2^{<\kappa}$. Then $\mathcal{W}(\mathbf{No}_{<\kappa})$ has cardinality 2^λ , is universal for linear orders of cardinality at most λ , and admits exactly the ordinals below λ^+ . The same conclusions hold for its cardinal-length permutation suborder and for the injective-word order $\mathcal{I}(\mathbf{No}_{<\kappa})$.*

The parameter controlling this spectrum is the cardinality λ of the alphabet. It need not equal the birthday cutoff κ . For $\kappa = \omega$, the alphabet consists of the dyadic rationals. Its well-ordering order has cardinality $2^{\aleph_0}$, contains every countable linear order, and does not contain ω_1 . For $\kappa = \omega_1$, the alphabet consists of the surreals with countable sign expansions and has cardinality $\mathfrak{c} = 2^{\aleph_0}$. Its well-ordering order consequently has cardinality $2^{\mathfrak{c}}$ and ordinal spectrum $\alpha < \mathfrak{c}^+$, without any use of the continuum hypothesis. When $2^{<\kappa} = \kappa$, the spectrum simplifies to $\alpha < \kappa^+$.

Lemma 7.2. *If κ is regular infinite, $\mathbf{No}_{<\kappa}$ fills every cut $A < B$ with $|A \cup B| < \kappa$. Such a cut can still be filled after excluding any prescribed subset of cardinality less than κ .*

Proof. The birthdays of the members of $A \cup B$ have supremum below κ , by regularity. The simplest surreal between them has birthday at most the supremum of their birthdays plus one, and hence still belongs to $\mathbf{No}_{<\kappa}$. For the exclusion statement, let S be the forbidden set, and let S_0 consist of its members that already lie strictly between A and B . Fill the cut $A \cup S_0 < B$. Its filler lies in the original interval but strictly above every forbidden point in that interval. $\square$

We call this property η_κ , always allowing either side of a cut to be empty. The cardinal parameter in this notation is explicit; conventions indexing eta orders by ordinal subscripts vary in the literature. This elementary form of saturation is all that the stage arguments below require.

Regularity is necessary for this particular cutoff conclusion. If κ is singular, a cofinal sequence of ordinals below κ , of length $\text{cf}(\kappa) < \kappa$, is unbounded in $\mathbf{No}_{<\kappa}$. Indeed, a surreal of birthday β is at most the ordinal β , the all-positive sign expansion of that length. Every candidate upper bound is therefore exceeded by some member of the cofinal sequence. The cut consisting of this sequence on the left and the empty set on the right has no filler in $\mathbf{No}_{<\kappa}$. This failure concerns the bounded stage; it does not weaken the set-cut theorem for the full surreal class.

[merge] The next section is source 12's Section 10. Its dichotomy is new against the repository; it answers in part the reals report's question "Other ground orders" (section 1.7). It uses source 12's self-delimiting code (lemma 18.16, printed in part IV) and its lemmas of section 6.2. Here S_κ of source 12 is printed $\mathbf{No}_{<\kappa}$, and its "binary coding rank" is the binary coding length ℓ_2 of the reals report.

8 Birthday cutoffs and the singular transition (source 12)

Let κ be an infinite initial cardinal and put

$$\mathbf{No}_{<\kappa} = \{s \in \mathbf{No} : \text{bd}(s) < \kappa\}, \quad \mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa} = \sup_{\lambda < \kappa} 2^{|\lambda|}.$$

The cardinality identity follows by counting all levels. The supremum is at least κ , and summing κ levels of size at most that supremum does not increase it. The reverse inequality is supplied by the levels themselves. We do not assume $\mathbf{No}_{<\kappa}$ is a field; none of the following proofs uses arithmetic closure.

8.1 A scheduler that preserves the first varying coordinate

Lemma 8.1 (Row scheduler). *Suppose a set order X of infinite cardinality κ has disjoint subsets $(X_i)_{i < \kappa}$, each containing a copy of X and each of cardinality κ . Suppose their complement is either empty or has cardinality κ . Then*

$$X_{\text{lex}}^\kappa \hookrightarrow \mathcal{W}_\kappa(X).$$

Proof. Fix embeddings $\phi_i : X \rightarrow X_i$ and a baseline enumeration of each X_i in length κ . For $x \in X$, transpose the label $\phi_i(x)$ with the first label of that baseline, giving an enumeration $r_{i,x}$ of X_i whose first value is $\phi_i(x)$. Its dependence on x is fixed and uniform. If the complement is nonempty, add a fixed filler row indexed before all variable rows. The row index order still has type $1 + \kappa = \kappa$.

Well-order row-stage pairs $(i, j) \in \kappa \times \kappa$ first by $\max(i, j)$ and then lexicographically within a level. This schedule has type κ : the predecessors of any pair have cardinality less than κ , since both coordinates there are bounded below κ , while the entire set of pairs has size κ . The first appearance of row i is $(i, 0)$, and these first appearances occur in increasing row order.

For an input $(x_i)_{i < \kappa}$, output the j th entry of r_{i,x_i} when pair (i, j) is scheduled. The output is an exhaustive bijection in type κ . If two inputs first differ in row i , every earlier row agrees, no later row has appeared before $(i, 0)$, and the first output discrepancy is $\phi_i(x_i)$ versus $\phi_i(y_i)$. Its sign is exactly the sign of the input comparison. $\square$

Lemma 8.2 (Surreal cone decomposition). *If $|\mathbf{No}_{<\kappa}| = \kappa$, then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa.$$

Proof. One embedding is the inclusion of bijective sequences among all sequences. For the other, partition the sign sequences with a minus into the cones with prefixes

$$p_i = +^i \frown -, \quad i < \kappa.$$

They are disjoint, and prefixing by p_i gives a numerical order isomorphism from $\mathbf{No}_{<\kappa}$ onto that cone. For $i, \alpha < \kappa$, $i + 1 + \alpha < \kappa$; this uses that κ is an initial cardinal and is valid even when it is singular. Removing the initial prefix also leaves a sign sequence of length less than κ . The complement consists of all-plus sequences, one for each length below κ , so has cardinality κ . Apply lemma 8.1. $\square$

Theorem 8.3 (Exact minimal-slice dichotomy). *For κ and $\mu = 2^{<\kappa}$ as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Consequently its binary coding rank is respectively $\kappa \cdot \kappa$ or μ . The exceptional case is exactly the case of a singular strong-limit cardinal κ .

Proof. We separate the three mechanisms.

Case 1: $\mu > \kappa$. By lemma 18.16, each member of $\mathbf{No}_{<\kappa}$ has a prefix-free numerical binary code of length less than κ . Pad each such code to length κ . Concatenating the entry codes of a μ -long enumeration produces a binary word of length $\kappa \cdot \mu = \mu$. The equality holds because μ is an initial cardinal strictly larger than κ : every proper partial product $\kappa \cdot \xi$, $\xi < \mu$, has cardinality below μ . Thus $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \hookrightarrow \mathbb{B}_\mu$. Pair coding gives the reverse embedding.

Case 2: $\mu = \kappa$ and κ is regular. Use the unpadding, self-delimiting code $c(s)$. In a sequence of κ entries, the sum of code lengths before any stage $\xi < \kappa$ is less than κ , by regularity. The full sum is κ , since every length is positive. Consequently concatenation embeds even $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$ into $\mathbb{B}_\kappa$. Pair coding again supplies the reverse embedding for $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$.

Case 3: $\mu = \kappa$ and $\delta = \text{cf } \kappa < \kappa$. The padded single-entry code embeds $\mathbf{No}_{<\kappa}$ into $\mathbb{B}_\kappa$, whence

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \hookrightarrow (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa \hookrightarrow \mathbb{B}_{\kappa \cdot \kappa}.$$

Choose increasing nonzero ordinals $(\lambda_j)_{j < \delta}$ cofinal in κ . Their ordinal sum is κ . As $\delta \cdot \kappa = \kappa$, partition the κ input coordinates into κ consecutive blocks of length δ . In each block, restrict its j th coordinate to sign sequences of exactly length λ_j . The lexicographic product of those coordinates is the binary cube of the sum of their lengths. Thus, over all the blocks, this restricted product is

$$\mathbb{B}_{\sum_{\xi < \kappa} \sum_{j < \delta} \lambda_j} = \mathbb{B}_{\kappa \cdot \kappa}.$$

It embeds into $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$, which embeds into $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ by lemma 8.2. This gives the claimed equimorphism. The rank assertions follow from lemma 6.4.

Finally, a singular strong limit satisfies $2^{<\kappa} = \kappa$. Conversely, suppose κ is singular and $2^{<\kappa} = \kappa$. If $2^\lambda = \kappa$ for some $\lambda < \kappa$, choose $\rho < \kappa$ with $\rho \geq \lambda, \text{cf } \kappa$. Then $\kappa \leq 2^\rho \leq 2^{<\kappa} = \kappa$, so $2^\rho = \kappa$, contradicting König's inequality $\text{cf}(2^\rho) > \rho$. Thus $2^\lambda < \kappa$ for every $\lambda < \kappa$. $\square$

Corollary 8.4 (A complete cube test for the minimal slice). *Let $\rho = \kappa \cdot \kappa$ in the exceptional case of theorem 8.3, and let $\rho = \mu$ otherwise. Then*

$$\mathbb{B}_\gamma \hookrightarrow \mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \iff \gamma \leq \rho.$$

In every case its ordinal embedding spectrum is still precisely $\{\beta : \beta < \mu^+\}$.

Example 8.5 (An unconditional singular example). Let $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$. It is a singular strong-limit cardinal of cofinality ω , independently of GCH. Thus

$$\ell_2(\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})) = \kappa \cdot \kappa > \kappa, \quad \ell_2(\mathcal{W}(\mathbf{No}_{<\kappa})) = \kappa^+.$$

All three displayed ordinals must be distinguished. The first two have the same cardinality, but are different coding lengths; the successor cardinal is larger still.

Example 8.6 (Regular cutoffs). For $\kappa = \omega$, $\mathbf{No}_{<\kappa}$ is the order of finite-birthday surreals, and the minimal slice is equimorphic with $\mathbb{B}_\omega$. For $\kappa = \omega_1$, its cardinality is $2^{\aleph_0}$, and the minimal slice is equimorphic with $\mathbb{B}_{2^{\aleph_0}}$, whether or not CH holds. At a strongly inaccessible κ , it is equimorphic with $\mathbb{B}_\kappa$.

8.2 Why the singular phenomenon is genuinely ordinal

A set of κ coordinates does not license arbitrary reordering of those coordinates in a lexicographic product. In the singular case, a short cofinal block of coordinate lengths can already consume ordinal length κ ; repeating that event κ times consumes $\kappa \cdot \kappa$. Replacing this ordinal sum by its cardinality would incorrectly collapse the rank back to κ . The strict-cube lemma shows that the lost information is mathematically detectable.

The row scheduler addresses a separate possible error: an arbitrary bijection $\kappa \times \kappa \rightarrow \kappa$ need not preserve the first varying row. Bounding by the maximum coordinate ensures that all earlier observable output is independent of later varying rows. This is why the product-to-permutation embedding is valid even at singular κ .

[merge] The last sentence of [corollary 8.4](#) is [corollary 7.1](#) for the minimal slice. The dichotomy computes binary coding length only; source 11’s question on the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$ without $2^{<\kappa} = \kappa$ ([research question 22.7](#)) stays open.

Part II

Set-length words

9 Set-length surreal words embed explicitly in the surreal line

Let $\mathcal{T}$ be the class of all set-length words $f : \alpha \rightarrow \mathbf{No}$, and let $\mathcal{I}$ be its injective subclass. The latter codes well-orders of set-sized subsets of $\mathbf{No}$. Both are ordered lexicographically with proper prefixes smaller. Their members are sets, so these are ordinary definable classes in GB, not collections whose members are proper classes. They should not be confused with all global well-orders of the entire surreal class.

In particular, a word cannot exhaust $\mathbf{No}$: its range is a set by Replacement. Passing from fixed alphabets to $\mathcal{I}$ consequently changes which carrier is being well-ordered from one point of the lexicographic order to another. This is a mathematically natural extension, but it answers a different question from ordering all class well-orders on a single proper-class carrier. Making this distinction before forming any collection of codes is what prevents a hidden ascent from classes to collections of classes.

The following construction uses neither a global enumeration of $\mathbf{No}$ nor Global Choice. It uses only ordinal concatenation of sign expansions, whose general properties are discussed in [\[14, Section 3.2\]](#).

Theorem 9.1 (Explicit sign coding). *There is an explicitly defined order embedding $\mathcal{E} : \mathcal{T} \rightarrow \mathbf{No}$. It maps the empty word to 0, maps every nonempty word to a positive surreal, and restricts to an order embedding of $\mathcal{I}$. Singleton words give embeddings in the opposite direction.*

Proof. For a sign expansion s , let $D(s)$ duplicate each sign: replace $-$ by $(-, -)$, and $+$ by $(+, +)$. For $x \in \mathbf{No}$, put

$$C(x) = (+) \frown D(s_x) \frown (+, -).$$

The length here is the ordinal $1 + 2 \cdot \text{bd}(x) + 2$. If s_x, s_y first differ at a sign, their doubled blocks have the same comparison. If s_x is a proper prefix of s_y , the terminal block $(+, -)$ of $C(x)$ is compared with the next doubled block of $C(y)$. We have

$$(-, -) < (+, -) < (+, +).$$

This is precisely the comparison of a missing sign with a negative or positive continuation. Thus C preserves the surreal order. Moreover, two distinct values of C cannot be prefixes of one another: the same first differing block supplies an actual differing sign before either code ends.

For $f : \alpha \rightarrow \mathbf{No}$, concatenate these blocks in domain order:

$$\mathcal{E}(f) = \bigwedge_{i < \alpha} C(f(i)).$$

Replacement and ordinal summation make this a set-length sign expansion. If two words first differ at i , their preceding blocks coincide and their blocks at i have a first differing sign. Prefix freedom prevents that difference from being displaced into a subsequent block. The comparison therefore agrees with the comparison of the original entries. If f is a proper prefix of g , the expansion $\mathcal{E}(g)$ continues $\mathcal{E}(f)$ with the initial $+$ of the next block, so $\mathcal{E}(f) < \mathcal{E}(g)$. This also proves the assertions about empty and nonempty words. Finally, $x \mapsto \langle x \rangle$ embeds $\mathbf{No}$ into both word classes. $\square$

Accordingly, $\mathcal{T}$, $\mathcal{I}$, and $\mathbf{No}$ are pairwise equimorphic as class linear orders. This conclusion asserts embeddings both ways, not isomorphism. Section 10 exhibits an obstruction to isomorphism that persists after the empty word is removed.

Proposition 9.2 (Birthday bounds). *The coding has the following bounds.*

1. *If κ is regular infinite, $\alpha < \kappa$, and $f(i) \in \mathbf{No}_{<\kappa}$ for every $i < \alpha$, then $\mathcal{E}(f) \in \mathbf{No}_{<\kappa}$.*
2. *For any infinite cardinal κ , put $\lambda = 2^{<\kappa}$. If f is an arbitrary injective word over $\mathbf{No}_{<\kappa}$, then $\mathcal{E}(f) \in \mathbf{No}_{<\lambda^+}$.*

Proof. The birthday of the image is

$$\text{bd}(\mathcal{E}(f)) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2),$$

with ordinal addition and multiplication. For the first assertion, this sums fewer than κ ordinals below κ , and regularity keeps the sum below κ . For the second, injectivity gives $|\alpha| \leq \lambda$; each block has cardinality at most λ . The whole sign expansion has cardinality at most λ , so its ordinal length is below λ^+ . $\square$

The first bound need not hold at a singular cutoff. Choose distinct ordinals $\gamma_i < \kappa$, $i < \text{cf}(\kappa)$, cofinal in κ , and regard them as surreal entries. The word has length less than κ , but the lengths of its code blocks are unbounded in κ , so its encoded birthday is at least κ . Also, injectivity is essential to the unrestricted second bound: constant words over a one-element alphabet can have every ordinal length.

10 Local characters and set suprema of word orders

In this section $\mathcal{W}$ denotes either $\mathcal{T}$ or $\mathcal{I}$. Deleting the set of entries of an existing prefix causes no difficulty when choosing a new coordinate: apply Lemma 5.2. We say that a segment has no set cofinal subset rather than assigning it a cardinal cofinality that does not exist.

Theorem 10.1 (Exact local character). *Let $f \in \mathcal{W}$ have domain α .*

1. *If α is a nonzero limit ordinal, then $\text{cf}(\mathcal{W}_{<f}) = \text{cf}(\alpha)$.*
2. *If α is a successor ordinal, then $\mathcal{W}_{<f}$ has no set cofinal subset.*
3. *For every α , the segment $\mathcal{W}_{>f}$ has no set coinitial subset.*

The empty word is the minimum of $\mathcal{W}$.

Proof. Suppose first that α is a nonzero limit. Its proper prefixes $f \restriction \beta$, $\beta < \alpha$, are cofinal below f . A word $g < f$ is either a proper prefix, or differs from f first at some coordinate ξ ; in the latter case it lies below every prefix long enough to include coordinate ξ . A cofinal sequence of indices in α therefore supplies a cofinal subset of size $\text{cf}(\alpha)$. Conversely, to each $g < f$ assign a prefix length strictly large enough that the corresponding prefix exceeds g . Fewer than $\text{cf}(\alpha)$ such lengths are bounded below α , so the corresponding words cannot be cofinal.

Next let $\alpha = \beta + 1$, write $p = f \restriction \beta$, and take a set $A < f$. For each member a properly extending p , its next coordinate satisfies $a(\beta) < f(\beta)$. Choose an unused surreal d strictly above all these coordinates and strictly below $f(\beta)$. Then $h = p \frown \langle d \rangle$ is below f . It exceeds every member of A extending p ; the remaining members are at most p , so they also lie below h . Thus A is not cofinal.

Finally, take a set $B > f$. Collect $b(\alpha)$ for those $b \in B$ properly extending f , and choose an unused d below all these coordinates. The word $h = f \frown \langle d \rangle$ lies above f . It lies below the members of B extending f . Every other member of B first differs earlier with a larger coordinate, and consequently exceeds every extension of f , including h . Hence $f < h < B$. $\square$

Corollary 10.2. *Neither $\mathcal{T}$ nor $\mathcal{I}$, with or without its empty word, is order-isomorphic to $\mathbf{No}$. Each has nontrivial set suprema, but a set with no minimum cannot have an infimum in either word order.*

Proof. Take an injective word f of length ω . Its nonempty finite prefixes increase to f , and f is their supremum by the preceding theorem. No surreal can be the supremum of a set lying strictly below it: the surreal set-cut property would insert a smaller upper bound. This distinguishes the orders even after deleting their minimum. If f were the infimum of a set B not containing f , part (3) would produce h with $f < h < B$, contradicting the definition of infimum. $\square$

These local obstructions coexist with very rich intervals. Given $f <_{\text{lex}} g$ with a first differing coordinate, vary that coordinate through a surreal interval strictly between the two values and disjoint from the set of preceding entries. If f is a proper prefix of g , vary a new coordinate below the next entry of g instead. The surreal set-cut property provides such an interval after excluding the preceding entries. Every surreal interval (u, v) is order-isomorphic to $\mathbf{No}$, for example by

$$t \mapsto u + \frac{v - u}{2} \left(1 + \frac{t}{1 + |t|} \right).$$

Thus every open word interval contains a copy of $\mathbf{No}$, and Theorem 9.1 then places a copy of the entire word order inside it. Interval universality therefore does not imply that set cuts are filled.

11 The complete obstruction to filling a set cut

The preceding prefix example is the only nontrivial obstruction. Here and below, $A < B$ means that every member of A is strictly below every member of B ; either set may be empty.

Theorem 11.1 (Word-cut classification). *A set cut $A < B$ in $\mathcal{W}$ has no interpolant if and only if one of the following holds:*

1. *B contains the empty word;*
2. *B has a minimum b of nonzero limit length, and A is cofinal in $\mathcal{W}_{<b}$.*

If neither obstruction occurs, an interpolant can be chosen to have nonzero successor length.

Proof. Either displayed obstruction plainly prevents an interpolant. For the converse, perform the following common-prefix search. Start with p empty. Maintain these invariants: members of A not extending p lie below p ; members of B not extending p lie above every extension of p . Here extending allows equality.

If $p \in B$, stop. Otherwise, with $\delta = \text{dom}(p)$, form the sets of next coordinates

$$A(p) = \{a(\delta) : a \in A, p \text{ is a proper prefix of } a\}, \quad B(p) = \{b(\delta) : b \in B, p \text{ is a proper prefix of } b\}.$$

Every member of $A(p)$ is at most every member of $B(p)$, since the corresponding words satisfy $a < b$. If these coordinate sets are disjoint, all these inequalities are strict. Choose a surreal c strictly between them, outside $\text{ran}(p)$. Then $p \frown \langle c \rangle$ interpolates the original cut by the invariants, and the construction is complete.

If the two coordinate sets meet, their intersection consists of a unique element c : two distinct common elements would violate the preceding inequalities. Replace p by $p \frown \langle c \rangle$. The invariants persist because the discarded A -branches have smaller next coordinates and the discarded B -branches have larger ones. In the injective-word case, c is unused, since it is witnessed by an injective word extending the previous prefix. At a limit stage take the union of the prefixes. It is a set word, is injective when required, and satisfies the same invariants. A member of B cannot have become a proper prefix of this union: the search would already have stopped upon reaching that member.

This procedure requires only set recursion. Choose an ordinal strictly above the domains of all words in $A \cup B$. The search cannot continue taking common next coordinates up to that length, because there would be no words supplying those coordinates. It therefore terminates. The remaining possibility is termination at $p \in B$. The invariants show that $p = \min B$. If its length is zero, the first obstruction holds.

Termination cannot occur at a successor length $\delta + 1$. The last appended coordinate was chosen from the intersection, so some member of A extends this successor prefix. Such a word is at least p , contrary to $A < B$. Hence a nonzero terminal length α is a limit. For every $\beta < \alpha$, a member of A witnessed the common coordinate at stage β ; it extends $p \upharpoonright (\beta + 1)$ and lies below p . These witnesses exceed prefixes of arbitrarily large length. Since the proper prefixes are cofinal below p , A is cofinal below p . This is exactly the second obstruction. $\square$

Corollary 11.2 (Successor words). *The suborder of $\mathcal{W}$ consisting of words of nonzero successor length fills every set cut. In GBC, it is order-isomorphic to **No**.*

Proof. Neither obstruction in Theorem 11.1 can occur when the upper set consists of successor words. The construction returns another successor word, proving the set-cut property. Under Global Choice, enumerate each of the two proper classes by a set-like class well-order and perform back-and-forth. At every ordinal stage the previously matched elements form a set; their images specify a set cut,

and an unused filler extends the partial order isomorphism. The union is an isomorphism. This is the class analogue of the usual eta-order back-and-forth, with its choice assumption made explicit; compare the class framework in [15]. $\square$

The distinction is precise. Deleting the empty word alone does not remove the prefix-supremum obstruction. Restricting to successor lengths does remove it. The explicit embeddings of Theorem 9.1 require no Global Choice, while the back-and-forth identification just given uses it.

12 Saturation at a regular stage

For regular infinite κ , restrict to successor-length injective words of length below κ , with entries in $\mathbf{No}_{<\kappa}$. This order is η_κ . Indeed, a cut with fewer than κ words has all its word lengths bounded below κ . The common-prefix search in the preceding proof consequently stops below κ , and Lemma 7.2 supplies the last coordinate while excluding the fewer than κ entries already used. Proposition 9.2 embeds this order into $\mathbf{No}_{<\kappa}$; singleton words embed $\mathbf{No}_{<\kappa}$ back. When $2^{<\kappa} = \kappa$, both orders have cardinality κ , and the ordinary back-and-forth of length κ makes them isomorphic.

A related saturation statement applies to full permutations.

Proposition 12.1. *Let X be an η_κ -order of cardinality κ , where κ is regular infinite. Then $\mathcal{W}_\kappa(X)$ is η_κ . Its global and local cofinalities and coinitalities are all exactly κ .*

Proof. For a cut $A < B$ of total cardinality below κ , apply the common-coordinate search to its permutations. If both sides are nonempty, the first-disagreement indices of the pairs in $A \times B$ have supremum below κ . The search cannot continue sharing a coordinate past this bound; if either active side becomes empty, it stops sooner. The common prefix therefore has length below κ . The η_κ property chooses an unused coordinate strictly between the active coordinate sets. Complete the resulting prefix to a κ -permutation: its remaining entries and remaining positions both have cardinality κ , and its remaining interval of positions has order type κ . The completed permutation fills the cut.

Saturation excludes cofinal or cointial subsets of size below κ , both globally and on either side of a point. Conversely, Theorem 6.1 excludes monotone sequences of length κ^+ . If one of these segments had no cofinal subset of size at most κ , recursively choosing a point above all earlier choices would produce such an increasing sequence. The decreasing argument is identical. Thus each of the stated characters equals κ . $\square$

In particular this proposition applies to $\mathbf{No}_{<\kappa}$ when $2^{<\kappa} = \kappa$. The resulting permutation order has cardinality 2^κ , so it is substantially larger than its alphabet while retaining the same local characters. It is not η_{κ^+} : that stronger cut property would construct an increasing κ^+ -sequence, contrary to Theorem 6.1.

Part III

Global choice and all class well-orders

13 Global choice and well-ordering the surreal class

The distinction between a set of surreal numbers and the entire surreal class changes the existence question. Here GB has the sets of ZF, without a choice axiom, and GC asserts the existence of a

class function choosing an element of every nonempty set. A class well-order is a strict linear class relation for which every nonempty available subclass has a least element. Its underlying objects are still sets. The relation itself can be a proper class, and its predecessor classes need not be sets. These are the size distinctions emphasized in the repository's foundations discussion [3].

Throughout this section, $\mathfrak{W}_{\text{all}}$, $\mathfrak{W}_{\text{sl}}$, and $\mathcal{W}(X)$ when X is a proper class are metanotation for the indicated collections of class relations. We quantify over their members as class variables; we do not assert that those collections are GB classes. When X is a set, $\mathcal{W}(X)$ retains its ordinary set meaning from the preceding section. All constructions below are uniform formulas in the stated class parameters. In particular, restricting such a construction to an already given uniformly indexed family produces an ordinary class relation on the indexing domain.

Concretely, a family indexed by a class I is supplied by one class $\mathcal{R} \subseteq I \times \mathbf{No} \times \mathbf{No}$, whose slice at i defines the relation R_i . Writing this family does not make R_i a set or a member of I : the index i is a set, and the slice is a class defined with that parameter. This distinction permits uniform families of proper class relations while preserving the usual GB typing rules. No exhaustive indexing family is assumed in this section.

Theorem 13.1 (Existence and global choice). *Over GB, the following statements are equivalent:*

1. GC holds;
2. there is a class well-order of $\mathbf{No}$;
3. there is a set-like class well-order of $\mathbf{No}$;
4. there is a class bijection $\text{Ord} \rightarrow \mathbf{No}$.

Here set-like means that the predecessors of every element form a set.

Proof. We give the implication from an arbitrary well-order to global choice in detail. It also explains why set choice need not be assumed in the base theory. Let W be a class well-order of the sign-sequence presentation of $\mathbf{No}$. By ordinary set-valued transfinite recursion, with the fixed class parameter W , construct a set relation w_α well-ordering V_α , for every ordinal α .

Set $w_0 = \emptyset$. Suppose w_β has been constructed. There is a unique ordinal θ and a unique order isomorphism

$$e : \theta \longrightarrow (V_\beta, w_\beta).$$

For $A \subseteq V_\beta$, let s_A be the sign sequence of length θ whose value at ξ is $+$ exactly when $e(\xi) \in A$. The map $A \mapsto s_A$ is a bijection between $\mathcal{P}(V_\beta)$ and the set of all sign sequences of length θ . Define

$$A w_{\beta+1} B \iff s_A W s_B.$$

This is a set relation by separation with a class parameter, and it well-orders $V_{\beta+1} = \mathcal{P}(V_\beta)$. No selection of an enumeration is involved: the enumeration of an already specified set well-order is unique.

At a nonzero limit λ , order V_λ first by rank, and within rank γ by the restriction of $w_{\gamma+1}$:

$$x w_\lambda y \iff \text{rank}(x) < \text{rank}(y) \quad \text{or} \quad (\text{rank}(x) = \text{rank}(y) = \gamma \quad \text{and} \quad x w_{\gamma+1} y).$$

Since λ is a limit, every required $\gamma + 1$ is below λ . A nonempty subset of V_λ has a least occurring rank; the specified well-order on that rank then supplies its least member. Thus the recursion continues. Its stages are sets, even though their union over all ordinals is a class. This is precisely the ordinary GB recursion theorem, rather than a recursion whose individual values are proper classes.

For a nonempty set A , every element of A belongs to $V_{\text{rank}(A)}$. Choose the $w_{\text{rank}(A)}$ -least element of A . The recursion and this prescription define a single class choice function, establishing **GC**.

Conversely, global choice implies choice for sets. For each ordinal α , the set of well-orders of the birthday layer $S_\alpha = \{-, +\}^\alpha$ is nonempty. Apply the global choice function to this set, and order **No** first by birthday and then by the selected order within the birthday layer. This is a class well-order. Every predecessor class is contained in $\bigcup_{\beta \leq \alpha} S_\beta$ for some α , hence is a set. This proves the set-like version directly.

Finally, for any set-like well-order R of the proper class **No**, assign to x the ordinal order type of its predecessor set. These ranks form an initial subclass of **Ord**. That subclass cannot be a set: otherwise the inverse rank map, by class replacement on a set domain, would make **No** a set. It is therefore all of **Ord**, and the inverse rank map is the required bijection. Conversely, a class bijection from **Ord** transports its well-order to a set-like well-order of **No**. $\square$

The connection between surreal enumerations and global choice is discussed by Hamkins [18]; the general equivalences for global choice are collected in [19]. The explicit rank recursion above proves the assertion over the stated choiceless GB base. It does not assert that a global choice class is definable without parameters in the first-order language of set theory. Definability is an additional requirement.

[merge] *Second proof over GB + AC (source 12)*. Source 12 proves the equivalence by coding transitive closures on ordinals; set choice is used only to code each $\text{TC}(\{x\})$ on an ordinal. Its equivalence includes non-set-like well-orders, as source 11's does. Its subsection on birthday order follows its theorem.

13.1 Source 12's route over GB with set choice

Theorem 13.2 (Choice equivalence). *Over GB plus set-level AC, the following are equivalent:*

- (i) *global choice;*
- (ii) *a class bijection $\text{Ord} \rightarrow \mathbf{No}$;*
- (iii) *a set-like class well-order of $\mathbf{No}$;*
- (iv) *a class well-order of $\mathbf{No}$, not necessarily set-like.*

Proof. Global choice supplies a set-like global well-order of the universe, and hence of the proper class **No**. A proper set-like well-order has type **Ord**, giving (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii). A bijection in (ii) pulls back ordinal order, giving (iii), and (iii) $\Rightarrow$ (iv) is immediate.

We prove the substantive reverse implication by coding sets. Suppose R well-orders **No**. Let x be a nonempty set and let $T_x = \text{TC}(\{x\})$; it contains x and all its elements. By set choice, there is a least initial ordinal λ equinumerous with T_x . There is a fixed definable ordinal τ_λ large enough to code a binary relation on λ , together with one distinguished index, by a binary sequence of length τ_λ . For example, concatenate a $\lambda \cdot \lambda$ -long relation table and a λ -long singleton marker.

Consider the *set* of codes for well-founded extensional relations on λ whose Mostowski collapse has range T_x , and whose marked index collapses to x . This set is nonempty. Convert 0, 1 to $-, +$, view the codes as surreals, and take its R -least member. Decode it, and choose the element of x represented by the least ordinal index that belongs to the marked node in the coded membership relation.

The result is an element of x , uniquely determined by x and the class parameter R . Elementary class comprehension gives the graph of this choice operation on all nonempty sets. It is a global choice function. Notice that only a set of codes for each x was minimized: set-likeness of R was not needed. $\square$

Remark 13.3 (The explicit set-choice hypothesis). The converse proof uses set choice to code T_x on an ordinal. We do not claim the same equivalence over GB with no set-level choice. A purported stronger choice equivalence must address that missing coding step.

13.1.1 Birthday is not a canonical global well-order

Ordering first by birthday and then by numerical value does not work. For $n < \omega$, let s_n be the length- ω sign sequence consisting of n minuses followed by pluses forever. Then

$$\text{bd}(s_n) = \omega, \quad s_{n+1} < s_n.$$

The whole descending sequence lies in one birthday level. Simplicity itself is well-founded but is only a partial order; two incompatible sign sequences need not be comparable by simplicity.

With global choice one may choose a well-order on every birthday level and concatenate them in birthday order. The result is set-like because all earlier levels, and every initial part of the current level, form a set. The noncanonical choices have not disappeared: they are exactly the kind of information accounted for by [theorem 13.2](#).

Remark 13.4 ([merge]; source 12's question answered). Dated note, 2 October 2026. Source 12's [remark 13.3](#) declines the choiceless case, and its research question on the exact weak-choice strength ([remark 22.15](#)) asks which principle is equivalent to a class well-order of the sign-sequence surreals over GB without set choice. Source 11's [theorem 13.1](#) answers it: global choice itself. The “missing coding step” that [remark 13.3](#) names is avoided, not supplied: the well-orders w_α of V_α are built by set-valued recursion from the given well-order of the sign sequences of length $\theta = \text{otp}(V_\beta, w_\beta)$, a subset $A \subseteq V_\beta$ being coded by the sign sequence s_A of length exactly θ , so no ordinal coding of a transitive closure has to be chosen. The proof thus produces a coherent sequence of well-orders of the sets V_α , in particular of the power sets of ordinals, which is the comparison the question suggests. The same mechanism, at the level of sets, is the ordinal-graph coding of the birthday-cutoff report (`hset:thm:choice`, whose range over ZF is the class of sets with well-orderable transitive closure) [4]; that theorem concerns a different statement and is unchanged. Hamkins's question whether surreal universality implies global choice [21] is not affected.

14 Lexicographic comparison without class ordinals

The set-sized enumeration method studied for the real numbers [1] suggests comparing the first unequal labels. Arbitrary class well-orders require a formulation that also permits a proper class of common earlier labels. No ordinal enumeration of that entire common prefix is needed.

For a strict class well-order R on a fixed class X , write

$$P_R(x) = \{y \in X : yRx\}.$$

A common labeled prefix of R, S is a subclass initial in both orders on which their restrictions agree. The adjective labeled matters: agreement here is equality of the actual elements and relations, rather than the existence of an abstract order isomorphism.

Lemma 14.1 (Unique first disagreement). *If $R \neq S$ are class well-orders of the same class X , there are unique distinct $a, b \in X$ such that*

$$P_R(a) = P_S(b) = C, \quad R \upharpoonright C = S \upharpoonright C.$$

The subclass C is their maximal common labeled prefix.

Proof. The class $D = \{x : P_R(x) \neq P_S(x)\}$ is nonempty, since predecessor classes determine a strict order. Let a be its R -least element and put $C = P_R(a)$. For every $y \in C$, $P_R(y) = P_S(y)$. In particular, ySa : otherwise aSy , which would imply aRy , contradicting yRa . Thus $C \subseteq P_S(a)$. Also, if zSy with $y \in C$, then zRy , so $z \in C$. Therefore C is S -initial as well as R -initial, and the restrictions of the orders to it agree.

The inclusion $C \subseteq P_S(a)$ is strict because $a \in D$. Let b be the S -least element outside C . The strict inclusion shows that bSa , hence $b \neq a$. Initiality and the minimality of b give $P_S(b) = C$.

For uniqueness, suppose a', b', C' satisfy the displayed conditions with $a' \neq b'$. Every element of C' has identical predecessor classes in the two orders. But $b'Sa'$, so the predecessor classes of a' differ. Consequently a' is exactly the R -least element of D ; hence $a' = a$, $C' = C$, and $b' = b$. Any common prefix properly extending C would have to include incompatible next labels, so C is maximal. $\square$

Fix now a native linear order $<_X$ on the labels. For distinct R, S , define $R <_{\text{lex}} S$ when the labels supplied by Lemma 14.1 satisfy $a <_X b$.

Theorem 14.2 (Raw lexicographic order). *This rule is a strict linear order on the class well-orders of X , understood as a definable relation on class variables. In particular it defines $\mathfrak{W}_{\text{all}}$ for $X = \mathbf{No}$ with its native numerical order.*

Proof. Existence and uniqueness of first disagreement give totality and asymmetry; the relation is irreflexive by definition. To verify transitivity, suppose $R <_{\text{lex}} S <_{\text{lex}} T$. Write (a, b) for the first unequal labels of R, S , and (c, d) for those of S, T . Their common prefixes are $C = P_S(b)$ and $D = P_S(c)$, respectively, and $a <_X b$, $c <_X d$.

If bSc , then $C \subsetneq D$. The orders S, T agree through the label b , so the first unequal labels of R, T are a, b , with common prefix C . Thus $R <_{\text{lex}} T$. If cSb , then $D \subsetneq C$; now R, S agree through c , and the first unequal labels of R, T are c, d , again giving the conclusion. If $b = c$, the prefixes coincide and the next labels of R, T are a, d . Since $a <_X b = c <_X d$, the conclusion follows in this case as well. $\square$

This proof compares labeled relations directly. It does not compare arbitrary class well-orders up to isomorphism, and makes no appeal to a collection of class ordinal types. The defining conditions use only set quantifiers with the given class relations as parameters: equality of predecessor classes means a universal statement about their set elements.

For example, the predicate that a, b are the first unequal labels can be written without quantifying over a common-prefix class:

$$\begin{aligned} a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \longrightarrow (xRy \leftrightarrow xSy)). \end{aligned}$$

Adding $a <_X b$ and existentially quantifying the two labels gives the lexicographic predicate. Consequently its restriction to the indexed family $\mathcal{R}$ described above is a GB class of pairs of indices. If the indexing contains duplicate orders, this relation orders the distinct relations, and becomes a strict linear order on indices whenever the indexing is injective. Passing instead to abstract isomorphism types would discard exactly the label information on which the construction depends.

Proposition 14.3 (Convex prefix cones). *Fix a well-ordered subclass (C, P) of X . The collection of all class well-orders of X extending (C, P) as an initial labeled segment is convex in the raw lexicographic order.*

Proof. Suppose R, S extend the prefix and $R <_{\text{lex}} T <_{\text{lex}} S$. Let D be the first-disagreement prefix of R, T . The subclasses C, D are initial in R , and therefore inclusion-comparable. If $C \subseteq D$, then T extends (C, P) , as required. If $D \subsetneq C$, the next R -label lies in C ; it is also the next S -label after D . Accordingly, T compares with R and S in the same direction, contradicting the strict inequalities. This reasoning permits C and D to be proper classes. $\square$

[merge] *Second route (source 12): the largest common labelled prefix.* Source 12's Section 3 proves lemma 14.1 and theorem 14.2 differently. Instead of the class D of points whose predecessor classes differ, it defines the largest common labelled prefix $C(R, S)$ by an elementary formula and proves directly that it is initial in both orders and contains every common labelled initial segment. Source 11's shipped check compares its predecessor rule with finite lexicographic order on all 266,272 pairs of permutations of at most six points (data/11-raw-orders-finite_checks_results.txt); source 12's checks the common-prefix formula on 15,017 finite pairs (data/12-singular-finite_checks.json).

14.1 Source 12's route: the largest common labelled prefix

The central observation is that comparing two *labelled* well-orders is easier than comparing arbitrary abstract class order types. Fix a class X equipped with an ambient strict linear order $<$, and let R, S be class well-orders of the same carrier X .

Definition 14.4 (Common labelled prefix). Define $C(R, S) \subseteq X$ by putting $x \in C(R, S)$ if

$$\begin{aligned} \text{Pred}_R(x) &= \text{Pred}_S(x), \\ \forall u, v \in \text{Pred}_R(x), \quad uRv &\iff uSv. \end{aligned}$$

Equality of the predecessor classes is expanded as a quantification over sets $u \in X$. Thus this is an elementary formula with class parameters R, S ; it is not a quantification over all possible common prefixes.

Lemma 14.5 (Maximal-prefix lemma). *$C(R, S)$ is an initial segment of both well-orders, their restrictions to it agree, and it contains every common labelled initial segment. If $R \neq S$, it is proper, and the least elements*

$$a = \min_R(X \setminus C(R, S)), \quad b = \min_S(X \setminus C(R, S))$$

exist and are distinct.

Proof. Suppose $x \in C(R, S)$ and yRx . Equality of the predecessor classes puts y before x in S as well. For zRy , transitivity gives zRx ; agreement of R, S below x then gives zSy . Conversely, zSy implies zSx , hence zRx , and agreement below x yields zRy . The orders agree below y because they agree below x . Therefore $y \in C(R, S)$. The symmetric argument proves initiality in S .

If $u, v \in C(R, S)$ and uRv , equality of the predecessor classes of v gives uSv . Thus the restrictions agree. If D is any common labelled initial segment and $x \in D$, both predecessor classes of x lie in D and agree, as do the restricted relations. Hence $D \subseteq C(R, S)$. If $C(R, S) = X$, equality of every predecessor class implies $R = S$.

For $R \neq S$, class well-foundedness supplies a, b . By initiality and minimality, $\text{Pred}_R(a) = C(R, S)$ and $\text{Pred}_S(b) = C(R, S)$. If $a = b$, these predecessor classes agree and the two relations agree on them. The defining formula would put a in $C(R, S)$, a contradiction. $\square$

Definition 14.6 (Full lexicographic comparison). For distinct R, S , put $R <_{\text{lex}} S$ precisely when $a < b$, with a, b from [lemma 14.5](#). Identical relations are equal in this order.

Theorem 14.7 (Full class lexicographic order). *In GBC, the preceding comparison is a strict linear ordering of the predicate of all class well-orders of a fixed linearly ordered class X . Its definition uses only elementary class comprehension. In particular, ETR and the comparability of abstract class well-order types are not required.*

Proof. Irreflexivity and trichotomy follow from the maximal-prefix lemma and trichotomy in X . For transitivity suppose $R <_{\text{lex}} S <_{\text{lex}} T$. The classes $C(R, S)$ and $C(S, T)$ are initial segments of the same well-order S , so they are comparable by inclusion.

If $C(R, S)$ is a proper subset of $C(S, T)$, the first S -element b after $C(R, S)$ belongs to $C(S, T)$. Thus T agrees with S through that element. The first disagreement of R, T is exactly the earlier R, S disagreement, with the same two labels $a < b$. Consequently $R <_{\text{lex}} T$. The opposite proper-inclusion case uses the earlier S, T disagreement and is identical in form.

If the two common prefixes are equal to C , let a, b, c be the first labels after C in R, S, T . The hypotheses give $a < b < c$. Hence R, T also first disagree after C , and $R <_{\text{lex}} T$. Every construction of C , and every assertion about its next elements, expands into set quantifiers with the given class relations as parameters. $\square$

Remark 14.8 (Why this does not prove class order-type comparability). In general one asks whether two arbitrary class well-orders admit an initial-segment isomorphism. ETR is a sufficient hypothesis for that principle [[27](#), Theorem 6]. Here no such isomorphism is constructed. Only their *identical labelled* portions are compared. As soon as the labels disagree, the fixed ambient order decides the answer. Confusing these two tasks would impose an unnecessary stronger axiom.

Example 14.9 (A disagreement after every ordinary ordinal). Fix $a < b$ in **No**. Well-order $\mathbf{No} \setminus \{a, b\}$ in type Ord. Append first a, b , and in a second order append b, a . The orders agree throughout their initial Ord-block and differ only after it. No ordinary ordinal indexes that first disagreement, but [theorem 14.7](#) compares them immediately: their common labelled prefix is $\mathbf{No} \setminus \{a, b\}$.

Corollary 14.10 (Agreement with enumeration lexicography). *For set-sized exhaustive ordinal enumerations, and for class bijections $\text{Ord} \rightarrow X$, the common-prefix definition agrees with first-difference lexicography.*

Proof. Before the first unequal entry, exactly the same labels occur in exactly the same order. Their ranges form the common labelled initial segment. No later label can belong to that segment by initiality and the distinct next labels. In the Ord case the disagreement is a nonempty class of ordinals; restricting it below any one witness gives its least member. $\square$

[merge] Source 12's [remark 14.8](#) is source 11's remark after [theorem 14.2](#) and in [section 19](#): comparability of abstract class well-orders by initial-segment isomorphisms follows from ETR [[27](#)], and neither labelled comparison needs it. Its [example 14.9](#) is source 11's [example 16.3](#), and its [corollary 14.10](#) is the corresponding sentence after [definition 18.1](#).

15 The canonical initial set-like part

Every class well-order of **No** has a canonical initial segment of type Ord, even when its entire order is not set-like. This gives a structural description of all well-orders using ordinary class functions.

Theorem 15.1 (Canonical initial part). *For $R \in \mathfrak{W}_{\text{all}}$, let*

$$I_R = \{x \in \mathbf{No} : P_R(x) \text{ is a set}\}.$$

Then I_R is a proper initial subclass of the domain or is the entire domain; in either case it is a proper class. Its induced order is set-like and has a unique order enumeration

$$F_R : \text{Ord} \longrightarrow I_R.$$

The order R is the concatenation of this enumeration with its induced well-order on $\mathbf{No} \setminus I_R$.

Proof. If $x \in I_R$ and yRx , then $P_R(y) \subseteq P_R(x)$, so separation makes $P_R(y)$ a set. Thus I_R is initial. Suppose it were a set. Since $\mathbf{No}$ is proper, there would be an R -least element x outside it. Every predecessor of x would then belong to I_R , and initiality would place every member of I_R before x . Hence $P_R(x) = I_R$, a contradiction.

The induced order on I_R has the same predecessor sets as R . The ordinal-rank map therefore identifies it with an initial subclass of Ord . As in Theorem 13.1, a set-sized range would make I_R a set by replacement. Its range is consequently all of Ord . The inverse map is unique, and initiality gives the asserted concatenation. $\square$

Example 15.2 (Two proper blocks). Under GBC, fix $e : \text{Ord} \rightarrow \mathbf{No}$ bijective. The disjoint classes $A = \{e(2\alpha) : \alpha \in \text{Ord}\}$ and $B = \{e(2\alpha + 1) : \alpha \in \text{Ord}\}$ partition $\mathbf{No}$, where the product is ordinal multiplication $2 \cdot \alpha$. Order each block by its displayed ordinal index and place all of A before all of B . The resulting well-order has class type $\text{Ord} + \text{Ord}$. Its canonical initial part is exactly A : a point of B has the whole proper class A among its predecessors. Hence an Ord -enumeration of the canonical part is not, in general, an enumeration of the entire well-order. Both blocks consist of ordinary surreal labels; their size affects their positions in the chosen well-order, not the status of individual surreal numbers as sets.

Let $\mathfrak{J}$ denote the metacollection of injective class functions $F : \text{Ord} \rightarrow \mathbf{No}$, with lexicographic comparison at their first ordinal of disagreement. Such a first disagreement exists for distinct functions: after choosing any witness, minimize within the set of earlier ordinals. Put $Y_F = \mathbf{No} \setminus \text{ran}(F)$.

Proposition 15.3 (Convex fibers). *Assume GBC. Every $F \in \mathfrak{J}$ occurs as F_R for some $R \in \mathfrak{W}_{\text{all}}$. The fiber over F is a convex copy of $\mathcal{W}(Y_F)$. Distinct fibers are ordered by the first unequal values of their functions F .*

Proof. The range of an injection from Ord is proper, since a set could not contain an injective image of all ordinals. Choose a well-order Q of Y_F , using the restriction of any fixed global well-order, and put the sequence F before Q . Each $F(\alpha)$ has exactly the set $F''\alpha$ of predecessors. Each residual point, however, has the proper class $\text{ran}(F)$ among its predecessors. Thus the resulting order has canonical function exactly F .

All orders with that function extend the same initial labeled order on $\text{ran}(F)$, and conversely every extension has that canonical function. Proposition 14.3 proves convexity. Restriction to Y_F and concatenation with F are inverse operations; first disagreement inside the residual orders proves that they preserve lexicographic comparison. Finally, if $F \neq G$, their first ordinal disagreement occurs before either residual order begins. It therefore determines the comparison of every member of one fiber with every member of the other. $\square$

We may summarize this result as the lexicographic-sum description

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in (\mathfrak{J}, <_{\text{lex}})} \mathcal{W}(Y_F).$$

This is uniform metanotation for the preceding theorem, not the assertion that its indexing hyperclass or all its fibers have been collected into a GB class. The set-like fragment $\mathfrak{W}_{\text{sl}}$ consists exactly of the fibers with $Y_F = \emptyset$, or equivalently the exhaustive enumerations. For finite Y_F of size m , its fiber is a convex block of $m!$ orders, with the usual lexicographic order on permutations.

Remark 15.4 (Singleton fibers and iteration). A singleton fiber need not be set-like. If F omits exactly one surreal z , its unique completion puts z after the entire Ord-sequence; it has class order type $\text{Ord} + 1$. Conversely, every exhaustive F gives a singleton set-like fiber. Extracting the initial part once also does not justify transfinite iteration of this operation through class-sized tails. Such an iteration would require a separate specification of its indexing order, its objects, and its recursion principle; none is asserted here.

16 All jumps arise from finite residual permutations

The full order has jumps despite the extreme density of the native surreal line. A proper common prefix can already have used every label between the two labels at which comparison is decided.

Lemma 16.1 (Extreme enumerations). *For a class Z with a native linear order, a class well-order is lexicographically greatest among all well-orders of Z exactly when it is the native reverse order. It is lexicographically least exactly when it is the native forward order.*

Proof. If a well-order places y before z with $y < z$, transpose these two labels throughout the order. The result is another class well-order. Its first changed label is z in place of y , so it is lexicographically greater. A greatest order must therefore be strictly decreasing in the native order, which determines the entire relation. Conversely, in the reverse native order the next label at any first disagreement is the greatest remaining label, so every different well-order is smaller. The least-order assertion follows by reversing the native order. The argument includes empty and singleton domains. $\square$

The lemma is an existence criterion as well as a characterization: a greatest enumeration exists only when the native reverse order itself is well-founded. It is therefore not permissible to use “list the remaining labels in decreasing order” as an unconditional completion rule. For example, on a copy of the natural numbers with its usual native order, the increasing enumeration is least, but there is no greatest well-order. Finite residual domains have both extrema, and the adjacency theorem shows that precisely such domains can support a jump after a common prefix.

Theorem 16.2 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be class well-orders of X . Let C be their first-disagreement prefix, let $a < b$ be their respective next labels, and put $Y = X \setminus C$. They are adjacent if and only if all three conditions hold:*

1. *there is no $y \in Y$ with $a < y < b$;*
2. *the R -tail on $Y \setminus \{a\}$ is the native reverse order;*
3. *the S -tail on $Y \setminus \{b\}$ is the native forward order.*

These conditions force Y to be finite. Thus adjacency is exactly ordinary finite permutation adjacency after a possibly proper class common prefix.

Proof. Suppose first that the pair is adjacent. A label $y \in Y$ strictly between a, b could be placed immediately after C , followed by the restriction of R to the other remaining labels. This would give a well-order strictly between R, S ; thus condition 1 holds.

Any lexicographically greater replacement of the R -tail after a would also give an intermediate order: its next label after C would still be $a < b$. Hence this tail is greatest, and Lemma 16.1 gives condition 2. A lexicographically smaller replacement of the S -tail after b similarly gives condition 3. These modifications use restrictions and finite transpositions of existing class relations; they require no new choice principle.

The overlap $Z = Y \setminus \{a, b\}$ is now well-ordered both forward and backward in the native order. If Z were not finite, repeatedly take its least element after deleting the finitely many already selected. Ordinary recursion on ω and replacement produce a set of strictly increasing selected elements. That set has no greatest member, contradicting reverse well-ordering. Thus Z , and hence Y , is finite. This argument does not assume that Z was a set in advance.

Conversely, suppose the three conditions hold and let $R <_{\text{lex}} T <_{\text{lex}} S$. Convexity of the prefix cone forces T to extend C . Let $t \in Y$ be its next label. A label below a makes $T < R$, and a label above b makes $T > S$; condition 1 excludes intermediate labels. If $t = a$, its remaining tail cannot exceed the greatest tail of R , so $T \leq R$. If $t = b$, its tail cannot precede the least tail of S , so $T \geq S$. Every possibility contradicts strict betweenness. $\square$

Example 16.3 (An adjacent pair of type $\text{Ord} + 2$). Assume GBC, choose surreal labels $a < b$, and give $C = \mathbf{No} \setminus \{a, b\}$ a set-like well-order P . Consider

$$R = P + (a, b), \quad S = P + (b, a).$$

Each has class order type $\text{Ord} + 2$, and R, S are adjacent. They have the same canonical initial enumeration, and constitute its entire two-point fiber. The notation $\text{Ord} + 2$ describes a class concatenation; it is not a set ordinal.

For $X = \mathbf{No}$, every adjacent pair has a cofinite, hence proper class, common prefix. Neither endpoint can be set-like. Moreover, if two set-like orders are given, their common prefix is a set. The deletion property of the surreal line leaves an unused label strictly between their next labels, and that set prefix can be completed to a set-like well-order. For completeness, this extension does not require an additional recursion principle. Retain the prescribed set prefix, put the new intermediate label next, and then append the restriction of a fixed set-like global well-order of $\mathbf{No}$ to the unused labels. A predecessor class in the appended part is contained in the union of the prescribed set prefix and a predecessor set of the fixed order, together with the one inserted label. It is therefore a set. Every label occurs once, and the concatenation is a well-order, so the completion belongs to $\mathfrak{W}_{\text{sl}}$. Thus the induced order $\mathfrak{W}_{\text{sl}}$ is dense, while $\mathfrak{W}_{\text{all}}$ has the finite terminal blocks classified above. Allowing all well-orders changes the local order structure in a concrete and unavoidable way.

[merge] *Second route (source 12): residual decomposition and adjacency.* Source 12's Section 8 proves theorem 16.2 in the form “the residual class is finite and the residual permutations are consecutive”; the two criteria are equivalent, since a tail that is both well-ordered and reverse well-ordered is finite. Its proposition 16.5 is the description of one prefix cone given by propositions 14.3 and 15.3. Its corollary 16.7 adds that interior points of the finite blocks with $n \geq 3$ are isolated in the order topology and that the set-like suborder is not dense in the full order, and its last paragraph adds the order-reversing involution induced by negation.

16.1 Source 12's route: unrestricted class well-orders and finite-tail geometry

The full order $\mathfrak{W}_{\text{all}}(\mathbf{No})$ contains $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ as an ordered subcollection, but it is not dense. The effect is not a failure of density of the surreal alphabet; it is exhaustion of almost all labels by a proper-class prefix.

Definition 16.4 (A prefix cylinder in the full order). Let $C \subsetneq X$ carry a fixed well-order P . Consider all class well-orders of X that have (C, P) as a labelled initial segment. Write this cylinder as $[P]_X$ and put $D = X \setminus C$.

Proposition 16.5 (Residual-order decomposition). *$[P]_X$ is convex and is order-isomorphic to the lexicographic order of all well-orders of the residual carrier D , with its inherited ambient order. If D is a set, this is the ordinary set order $\mathcal{W}(D)$, with the same notation extended to finite carriers. If $|D| = n < \omega$, the cylinder has exactly $n!$ elements.*

Proof. Restriction to D and ordered concatenation with P are inverse operations. Concatenation of two class well-orders is a class well-order: a nonempty subclass has its minimum in the first part if it meets that part, and otherwise in the second. The maximal common prefix after P is exactly the common prefix of the tails, so comparisons are preserved.

If a third order differs before completing P , its first discrepancy with every extension of P is the same and places it on the same side of all of them. This proves convexity. The last assertion counts permutations of a finite carrier. $\square$

Theorem 16.6 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be two class well-orders of X , and let $C = C(R, S)$, $D = X \setminus C$. They are adjacent in the full order if and only if D is finite and their residual permutations are consecutive in the ordinary finite lexicographic order.*

Proof. Let $a < b$ be their first residual labels. If some $d \in D$ satisfies $a < d < b$, begin the residual order with d and well-order the remaining labels by a restriction of R . This gives an order strictly between. So an adjacent pair can have no such d .

If the tail of R after a has two labels u before v with $u < v$ in the ambient order, interchange the labels u, v at their two positions. The new well-order is larger than R , and still smaller than S because its first residual label remains a . Thus adjacency forces the R tail to be numerically decreasing. Dually, the tail of S after b must be numerically increasing.

On $D \setminus \{a, b\}$, both the ambient numerical order and its reverse are therefore well-orders. Such a carrier is finite: an infinite well-order contains an increasing ω -sequence, contradicting well-foundedness of its reverse. Hence D is finite.

Conversely, if D is finite, convexity in [proposition 16.5](#) says that every order between R, S must be another permutation of D after C . There is none exactly when the two permutations are consecutive. $\square$

Corollary 16.7 (Proper-class prefixes are responsible). *Adjacent pairs in $\mathfrak{W}_{\text{all}}(\mathbf{No})$ share a proper-class prefix. There are convex blocks of every factorial size $n!$. For $n \geq 3$, interior elements of such blocks are isolated in the order topology. The set-like suborder is not dense in the full order.*

Proof. A set prefix cannot exhaust all but finitely many surreals. Conversely, well-order $\mathbf{No} \setminus D$ in type Ord for any finite D , and append all its permutations. For $|D| \geq 3$, choose two nonconsecutive permutations in the resulting finite convex block. Their nonempty open interval contains only orders with a proper-class predecessor segment, so it contains no set-like order. $\square$

The full order still has no endpoints and no set-indexed cofinal or cointial family, by the first-entry argument of [note 18.13](#). It contains every class linear order, by [note 18.10](#). But neither global density nor interpolation for all set-indexed cuts survives: an adjacent pair already gives a two-sided cut that cannot be filled.

Pointwise surreal negation gives an order-reversing involution on both global orders: transport a well-order by the bijection $x \mapsto -x$. This preserves enumeration type and set-likeness while reversing the ambient comparison of the first unequal labels.

17 Numerically well-ordered subclasses and other interpretations (source 12)

The positive surreal ordinals form a numerically increasing class copy of $\mathbf{Ord}$, but this is not a well-order of all of $\mathbf{No}$. More generally, [lemma 5.5](#) implies that every class well-order is isomorphic to a numerically well-ordered subclass of $\mathbf{No}$. That subclass need not be set-like in its inherited numerical order.

For an explicit example, the class

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \mathbf{Ord} \right\} \cup \{0\}$$

has numerical order type $\mathbf{Ord} + 1$. The arithmetic here is surreal arithmetic on the embedded ordinal values; the reciprocal reverses positive order and the minus sign reverses it back. The final point has a proper class of predecessors. Putting a positive ordinal block after this negative block gives a numerical copy of $\mathbf{Ord} + \mathbf{Ord}$.

For a set $A \subseteq \mathbf{No}$, numerical well-ordering has the usual ordinal order type. Its birthday bound places it inside some $\mathbf{No}_{<\kappa}$. The class of all such set-sized well-ordered subsets is a legitimate class of sets. Its members are not exhaustive enumerations of a fixed carrier, however, so one must specify what happens when one enumeration is a prefix of another. The “shorter first” convention and the surreal $- < \mathbf{end} < +$ convention are different choices; neither can be silently borrowed from the prefix-free exhaustive setting.

Finally, ordering binary relation tables rather than enumerations is another legitimate but different construction. If one fixes a set-like well-order of $\mathbf{No} \times \mathbf{No}$, compares the characteristic functions of relations at the first unequal table entry, and restricts to well-orders, one gets a table-dependent order. A first changed relation bit is not generally the first changed enumerated label. Even on three points the two lexicographic orders can differ. The present article uses labelled initial segments, not relation-table lexicography.

[merge] Source 12’s shipped check records a three-point example on which relation-table lexicography and enumeration lexicography differ (`data/12-singular-finite_checks.json`, key `relation_table_vs_enumeration_example`).

Part IV

Set-like global well-orders

18 The lexicographic structure of global enumerations

The set-like case has a particularly strong structure theorem. Although the full collection of global enumerations is a higher-order object, one ordinary class of set codes already meets every cut

specified by set-indexed families. That class, with its induced order, is isomorphic to the surreal line. The distinction between a set-indexed family and an ordinal-indexed class family is essential: an explicit omitted-range construction shows exactly where the interpolation theorem stops.

Throughout this section we work in **GBC** and fix a class bijection

$$e_0 : \text{Ord} \longrightarrow \mathbf{No}.$$

The existence of such a baseline is supplied by global choice; the relevant equivalences are discussed in [19]. We use the surreal set-cut property, its preservation under deleting a set (Lemma 5.2), and the universality and uniqueness statements of Proposition 5.4. These are order-theoretic properties; the field operations on **No** play no role in the main construction.

18.1 Class parameters and uniform families

Definition 18.1. For this section, $\mathfrak{J}$ denotes the collection of class injections $F : \text{Ord} \rightarrow \mathbf{No}$, and $\mathfrak{W}_{\text{sl}}$ denotes the subcollection of class bijections. For distinct $F, G \in \mathfrak{J}$, put

$$d(F, G) = \min\{\alpha \in \text{Ord} : F(\alpha) \neq G(\alpha)\}, \quad F <_{\text{lex}} G \iff F(d(F, G)) < G(d(F, G)).$$

A *uniformly set-indexed family* $(F_i)_{i \in I}$ is given by a set I and one class relation whose i th row is the graph of F_i .

These notations are interpreted relationally in two-sorted class theory. An individual F is a class, not a possible element of a GB class. Thus “for all $F \in \mathfrak{W}_{\text{sl}}$ ” abbreviates quantification over class functions with the stated property. Likewise, a uniformly indexed family is a single class of triples (i, α, x) , not a literal set containing proper classes. The first-difference definition is legitimate: choose any coordinate of disagreement and minimize within the set of preceding ordinals. The usual first-difference proof gives a strict linear comparison on these objects.

Uniformity matters whenever a set bound is taken. For example, if two uniformly set-indexed families have different rows across every pair, Class Replacement makes their cross first-difference ordinals a set. No principle of choosing an arbitrary family of classes is being assumed. Finite lists of given class parameters are automatically uniform, and the families of set-coded objects constructed below have a fixed evaluation formula, so their set-indexed subfamilies are uniform as well.

18.2 A class of bounded-support codes

Definition 18.2. A *normalized bounded-support code* is a set permutation $\sigma : \lambda \rightarrow \lambda$ of an ordinal λ satisfying

$$\lambda = \sup\{\xi + 1 : \xi < \lambda \text{ and } \sigma(\xi) \neq \xi\}.$$

The empty permutation is allowed. Extend σ to a class permutation $\hat{\sigma}$ of **Ord** by the identity outside λ , and evaluate the code as

$$E_\sigma = e_0 \circ \hat{\sigma}.$$

Write $\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{bd}}(e_0)$ for the ordinary class of these codes, ordered by the lexicographic comparison of their evaluations.

Normalization removes the otherwise harmless duplication obtained by adding fixed points to the end of a code. Every class permutation with bounded support has exactly one normalized code: restrict it to the supremum of one plus its moved coordinates. Its support is invariant under the permutation, so this restriction is a permutation of the indicated ordinal. Consequently distinct

normalized codes evaluate to distinct enumerations, and the induced order on $\mathcal{P}_{\text{bd}}$ is a genuine class linear order. The class is proper, since the transpositions of 0 with arbitrary nonzero ordinals have distinct codes.

Here and below, identifying a code with its evaluated enumeration is a notational convenience. It does not convert the evaluated proper-class graph into a set. In particular, the assertion that $\mathcal{P}_{\text{bd}}$ is an ordinary class refers to the codes, while assertions comparing its members with arbitrary members of $\mathfrak{M}_{\text{sl}}$ use the common evaluation relation.

Lemma 18.3 (Completion of an arbitrary set prefix). *Every injective set function $p : \delta \rightarrow \mathbf{No}$, with δ an ordinal, is the initial segment of an enumeration represented in $\mathcal{P}_{\text{bd}}$. The completion can be chosen uniformly in p .*

Proof. Let $q = e_0^{-1} \circ p$. This is a set injection from δ into Ord . Choose an infinite initial cardinal γ strictly larger than δ and than $\sup \text{ran}(q)$. Both complements

$$\gamma \setminus \delta \quad \text{and} \quad \gamma \setminus \text{ran}(q)$$

have cardinality γ , since fewer than γ elements were removed. A bijection between the complements therefore extends q to a permutation of γ . Extend that permutation by the identity above γ , and normalize its code. Its evaluation agrees with p at every coordinate below δ . To make the construction uniform, fix the least permissible γ and then the least complement bijection under a fixed global well-order. These choices select sets; they do not require a choice principle for families of proper classes. $\square$

The lemma is stronger than merely extending p to some global well-order. Even an arbitrarily long set-sized prefix can be realized by changing the baseline only on a bounded set of coordinates. The bound need not be small: it must contain the baseline indices of all values prescribed by p .

Source 12's prefix extension. [merge] Source 12's Section 6 ("Set-like global well-orders: interpolation and universality") proves the results of this section by the same arguments, except for its prefix extension, printed next, which is a different construction: it appends the restriction of the baseline order to the complement of the prefix, so its completion need not have bounded support. Source 12's interpolation theorem, local characters and local universality are recorded in notes below; its avoidance lemma and its universality lemmas are printed in [section 5](#).

We now restrict to $\mathfrak{M}_{\text{sl}}(\mathbf{No})$, the class-bijection predicate. All statements about set- or class-indexed collections of its members mean uniform families, unless a larger ambient universe is explicitly specified.

Lemma 18.4 (Extending a set prefix). *Every injective set sequence $p : \alpha \rightarrow \mathbf{No}$ extends to a class bijection $\text{Ord} \rightarrow \mathbf{No}$. The extension can be chosen uniformly from p and one fixed baseline bijection B .*

Proof. Let $A = \text{ran}(p)$. Restrict the baseline order to $\mathbf{No} \setminus A$. It is a proper-class set-like well-order, so it has its canonical increasing enumeration in type Ord . Append that enumeration to p . The class order $\alpha + \text{Ord}$ is canonically isomorphic to Ord . The construction is set-valued recursion with B as a parameter. $\square$

18.3 Interpolation of every uniformly set-indexed cut

Theorem 18.5 (Prefix interpolation). *Let $(A_i)_{i \in I}$ and $(B_j)_{j \in J}$ be uniformly set-indexed families in $\mathfrak{J}$, with*

$$A_i <_{\text{lex}} B_j \quad (i \in I, j \in J).$$

Either index set may be empty. There is an injective set prefix p such that every member of $\mathfrak{J}$ extending p is strictly above every A_i and strictly below every B_j . In particular, a member of $\mathcal{P}_{\text{bd}}$ lies between the two families.

Proof. If both index sets are nonempty, Class Replacement gives a set of ordinals $d(A_i, B_j)$. Choose β so that

$$d(A_i, B_j) < \beta \quad (i \in I, j \in J).$$

If one side is empty, put $\beta = 0$. We build an injective prefix p , keeping track of the rows that still agree with it. Call these rows *active*. The invariant is that every discarded A_i is below every extension of p , and every discarded B_j is above every extension of p .

Suppose p has length α and the construction has not stopped. Let

$$L = \{A_i(\alpha) : A_i \text{ is active}\}, \quad R = \{B_j(\alpha) : B_j \text{ is active}\}.$$

These are sets. Since active cross pairs agree before α , the hypothesis implies $\ell \leq r$ whenever $\ell \in L$ and $r \in R$. Thus $L \cap R$ has at most one member: two distinct common values, taken in reverse order as a left and a right value, would contradict this inequality.

If $L \cap R$ is empty, all cross inequalities are strict. By Lemma 5.2, choose

$$z \in \mathbf{No} \setminus \text{ran}(p), \quad L < z < R.$$

This notation includes an empty left or right side; if both are empty, choose any fresh value. Append z and stop. Every previously active left row is now strictly below every extension of the new prefix, and every previously active right row is strictly above it. The invariant handles the already discarded rows, proving the desired conclusion.

If $L \cap R = \{z\}$, append z and retain as active exactly the rows whose next value is z . The new value is fresh because each surviving row is injective. A discarded left value is strictly below z , since $z \in R$; a discarded right value is strictly above z , since $z \in L$. Hence the invariant is preserved. At a limit stage take the union of the preceding prefixes; its length is a set ordinal and it remains injective.

The procedure cannot continue with both sides active through a prefix of length β . Such active rows would agree below β , contrary to the bound on every cross first difference. Therefore it stops no later than the next step, giving a set prefix. Lemma 18.3 supplies a bounded-support completion. Notice that surjectivity of the input rows was never used; this is why the theorem was stated for $\mathfrak{J}$. $\square$

All recursion histories in this proof are sets. Even when the input rows are proper classes, their restrictions to a set ordinal are sets, and their next values form sets because the index sets are sets. The proof therefore uses ordinary set-valued transfinite recursion with class parameters, not ETR. The same observation applies to the back-and-forth argument below.

Note 18.6 (Source 12, Theorem 6.3, “Set-indexed interpolation”). *If I, J are sets and $(E_i)_{i \in I}, (F_j)_{j \in J}$ are uniform families in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} F_j$ for all i, j , there is $G \in \mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} G <_{\text{lex}} F_j$; empty sides are allowed, so $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ is dense without endpoints in the indexed-family sense. This is theorem 18.5, by the same forced-prefix construction; source 12 completes the prefix by lemma 18.4, so its interpolant need not lie in the bounded-support core.*

Corollary 18.7. *The class order $\mathcal{P}_{\text{bd}}$ is an η_{Ord} order and is order-isomorphic to $(\mathbf{No}, <)$. Moreover, it meets every strict cut in $\mathfrak{W}_{\text{sl}}$ specified by uniformly set-indexed families.*

Proof. A set of codes is uniformly evaluable, so Theorem 18.5 applies to every separated pair of sets in $\mathcal{P}_{\text{bd}}$. Empty sides give the absence of endpoints, and singleton sides give density. We already know that $\mathcal{P}_{\text{bd}}$ is a proper class. Proposition 5.4 therefore identifies it with $\mathbf{No}$.

Explicitly, global choice enumerates each of the two classes by ordinals. Alternately extend a set-sized partial isomorphism to cover the next element of one enumeration and then the next element of the other. The lower and upper images at either step are sets, so the cut property supplies the new point. The union of this ordinary set-valued recursion is an isomorphism. The last assertion is the direct specialization of Theorem 18.5 to bijective input rows. $\square$

This is an isomorphism of the class of bounded-support codes with the surreal line. It is not an isomorphism between $\mathbf{No}$ and the full higher-order collection $\mathfrak{W}_{\text{sl}}$. Different baselines produce classes with the same abstract order type, but the argument does not single out a canonical isomorphism.

Corollary 18.8 (The class of realizations of a set cut). *Under the hypotheses of Theorem 18.5, the ordinary class*

$$K = \{p \in \mathcal{P}_{\text{bd}} : (\forall i \in I \ A_i <_{\text{lex}} p) \text{ and } (\forall j \in J \ p <_{\text{lex}} B_j)\}$$

is itself order-isomorphic to $\mathbf{No}$.

Proof. It is an ordinary class by comprehension with the two uniform input relations as parameters. Suppose $C, D \subseteq K$ are sets with $C < D$. Add the evaluated codes of C to the lower input family and those of D to the upper input family. All cross inequalities remain strict: members of K already lie between the original sides. Theorem 18.5 supplies a member of K between C and D . Thus K has the set-cut property, including empty sides. It cannot be a set, since applying that property with lower side all of K would give a member above itself. The uniqueness statement of Proposition 5.4 now gives the isomorphism. $\square$

Thus the interpolation theorem produces a full proper-class order of available choices, even when the input constraints are numerous and occur at arbitrarily complicated set-ordinal coordinates. In particular, no set of codes can be order-dense in $\mathfrak{W}_{\text{sl}}$. Given such a set, choose a point strictly above all its evaluations, and then a second, larger point. The open interval between the two contains no code from the proposed dense set. The same argument applies within K : a set cannot be dense even in the class of realizations of one specified set cut. The ordinary class $\mathcal{P}_{\text{bd}}$ is therefore a natural scale for density here, while an exhaustive class of set codes will be ruled out by a separate diagonal argument.

18.4 Universality, cylinders, and local cofinality

Proposition 18.9. *The surreal line embeds explicitly into $\mathcal{P}_{\text{bd}}$. Consequently every ordinary class linear order embeds into $\mathcal{P}_{\text{bd}}$ and hence into $\mathfrak{W}_{\text{sl}}$, in the uniform evaluation sense.*

Proof. For $x \in \mathbf{No}$, transpose the values $e_0(0)$ and x in e_0 , and denote the resulting enumeration by E_x . Its coordinate permutation has support contained in $\{0, e_0^{-1}(x)\}$, so it has a code in $\mathcal{P}_{\text{bd}}$, and

$$E_x(0) = x, \quad x < y \iff E_x <_{\text{lex}} E_y.$$

For the general assertion, compose this embedding with the class-order universality of $\mathbf{No}$ from Proposition 5.4. $\square$

Thus all set ordinals, the class order Ord , its reverse, and class orders such as $\text{Ord} + \text{Ord}$ occur as suborders of the lexicographic comparison. This statement concerns the order *between* enumerations. Each individual enumeration in $\mathfrak{W}_{\text{sl}}$ still presents a well-order of type Ord ; its lexicographic environment need not be well-ordered.

Note 18.10 (Source 12, Theorem 6.7, “Local class universality and cylinders”). *Every class linear order has a uniform order-embedding into $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ and into every nonempty open interval of $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; for a set-sized injective prefix p , the cylinder $[p]$ is convex and isomorphic to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ by uniform transformations.* This is [proposition 18.9](#) and [proposition 18.11](#), by the same constructions ($x \mapsto E_x$ composed with [lemma 5.5](#); a cylinder through $\mathbf{No} \setminus \text{ran}(p) \cong \mathbf{No}$, [lemma 5.6](#)). Source 12 lists the examples “all set ordinals, their reverses, $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, all set-sized lexicographic powers of $\mathbf{No}$ formed as class orders” and adds that “every class linear order” still means an order whose elements are sets; it does not include an unrestricted higher collection such as all class well-orders themselves.

Proposition 18.11 (Prefix cylinders). *For an injective set prefix p , let*

$$C_p = \{E \in \mathfrak{W}_{\text{sl}} : E \upharpoonright \text{dom}(p) = p\}.$$

The cylinder C_p is convex and order-isomorphic to $\mathfrak{W}_{\text{sl}}$, with the isomorphism interpreted as a uniform transformation of class parameters. Every nonempty open interval in $\mathfrak{W}_{\text{sl}}$, and every interval specified by a uniformly set-indexed strict cut, contains a whole such cylinder.

Proof. An enumeration that first departs from p at a coordinate of p lies on the same side of all its extensions. This proves convexity. Put $S = \text{ran}(p)$ and $\delta = \text{dom}(p)$. By [Lemma 5.2](#) and [Proposition 5.4](#), fix an order-isomorphism $h : \mathbf{No} \rightarrow \mathbf{No} \setminus S$. The ordinal tail $[\delta, \text{Ord})$ has order type Ord , with its canonical increasing parametrization $\xi \mapsto \delta + \xi$. Given $E \in \mathfrak{W}_{\text{sl}}$, prepend p to the enumeration $h \circ E$ on this tail. The result is a bijection onto $\mathbf{No}$, since its prefix covers S and its tail covers exactly the complement. Conversely, restrict an element of C_p to its tail and apply h^{-1} . These transformations are inverse and preserve first-difference comparisons. The final assertion follows from [Theorem 18.5](#). $\square$

In particular, every such interval contains a copy of the entire comparison structure $\mathfrak{W}_{\text{sl}}$, not merely a copy of an arbitrarily large set order. This self-similarity does not invoke a class whose members are the cylinders; for each set prefix the displayed construction gives the required transformation, using the indicated class parameters.

Proposition 18.12 (Approximants at every point). *For each $F \in \mathfrak{J}$ there are uniformly ordinal-indexed families (a_α) and (b_α) of codes in $\mathcal{P}_{\text{bd}}$ such that, whenever $\alpha < \beta$,*

$$a_\alpha <_{\text{lex}} a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta <_{\text{lex}} b_\alpha.$$

The lower family is cofinal among the global enumerations below F , and the upper family is coinital among those above F . Neither side admits a uniformly set-indexed cofinal or coinital family, respectively.

Proof. Choose a fresh value below $F(\alpha)$, append it to $F \upharpoonright \alpha$, and use [Lemma 18.3](#) to obtain a_α . Choose a fresh value above $F(\alpha)$ to obtain b_α . [Lemma 5.2](#) supplies the fresh values, and the fixed global well-order makes all choices uniform. For $\alpha < \beta$, the comparison between a_α and a_β is decided at α , giving the asserted inequality; the other comparisons are identical arguments.

If $E <_{\text{lex}} F$, choose $\alpha > d(E, F)$. Then a_α agrees with F at the decisive coordinate, so $E <_{\text{lex}} a_\alpha <_{\text{lex}} F$. This proves cofinality, and the upper assertion is symmetric. Finally, given any

uniformly set-indexed family below F , Theorem 18.5 applied with upper side $\{F\}$ supplies a larger lower approximant in $\mathcal{P}_{\text{bd}}$. Hence the original family was not cofinal. The dual argument proves the coinitiality assertion. $\square$

For $F \in \mathfrak{W}_{\text{sl}}$, these are the local cofinality and coinitiality statements at an actual point of the order. It is appropriate to say that ordinal-length class families suffice and set-sized families do not; assigning an ordinary set cardinal to these proper-class cofinalities would obscure the distinction.

Note 18.13 (Source 12, Proposition 6.4, “Cofinality and local character”). *No set-indexed uniform family is cofinal or coinitial in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; there are increasing cofinal and decreasing coinitial families indexed by Ord; at every point neither side has a set-indexed cofinal, respectively coinitial, family, but both have Ord-indexed ones.* This is proposition 18.12 together with the remark that first entries of a set-indexed family are bounded, by the same argument; the cofinal family is $(E_\alpha)_{\alpha \in \text{Ord}}$ for the positive ordinals α .

Corollary 18.14 (Set saturation is not Dedekind completeness). *A nonempty uniformly set-indexed family in $\mathfrak{W}_{\text{sl}}$ with no greatest member has no supremum in $\mathfrak{W}_{\text{sl}}$. Dually, a family with no least member has no infimum. Both statements also hold in the ordinary class order $\mathcal{P}_{\text{bd}}$.*

Proof. Suppose that E is the supremum of a family (A_i) with no greatest member. Every A_i is then strictly below E . By Theorem 18.5, there is a bounded-support enumeration p with $A_i < p < E$ for every i . This makes p a smaller upper bound, a contradiction. The infimum argument reverses the inequalities, and the same bounded-support interpolant proves the assertions internal to $\mathcal{P}_{\text{bd}}$. $\square$

For example, take the enumerations E_n from the transposition construction, so that $E_n(0) = n$ for $n \in \mathbb{N}$. They form an increasing sequence of bounded-support codes. The enumeration whose first value is the surreal ordinal ω is an upper bound, but there is no least upper bound in either $\mathcal{P}_{\text{bd}}$ or $\mathfrak{W}_{\text{sl}}$. The obstruction is not a failure to find bounds: Theorem 18.5, with one input side empty, bounds every uniformly set-indexed family above and below. Rather, whenever a proposed bound is strictly beyond every member, there remains room to move the bound closer. This distinction will also matter when the full order of Dedekind cuts is embedded below. An order embedding preserves comparisons, but does not automatically preserve the role of a point as the least upper bound of a specified family.

18.5 Binary class words, cuts, and diagonalization

Theorem 18.15. *The lexicographic orders $\mathfrak{W}_{\text{sl}}$ and $\{0, 1\}^{\text{Ord}}$ are order-bi-embeddable, where a binary word of length Ord means a class characteristic function. Both embeddings have uniform relational definitions in GBC.*

Proof. Partition the coordinate ordinals into successive pairs $2 \cdot \xi, 2 \cdot \xi + 1$. Here the multiplication is ordinal multiplication with 2 on the left; every ordinal has a unique such pair position. Sort the corresponding baseline values as

$$u_\xi = \min\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}, \quad v_\xi = \max\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}.$$

For a class $D \subseteq \text{Ord}$, put these two values in the order (u_ξ, v_ξ) if $\xi \notin D$, and in the order (v_ξ, u_ξ) if $\xi \in D$. Denote the resulting enumeration by T_D . Each pair retains exactly its original two values, so T_D is a bijection onto $\mathbf{No}$. At the least differing bit of two characteristic functions, 0 chooses the smaller first value and 1 the larger one. Thus $D \mapsto T_D$ preserves and reflects lexicographic comparison.

For the converse, fix an ordinal enumeration $(p_\xi)_{\xi \in \text{Ord}}$ of the set codes in $\mathcal{P}_{\text{bd}}$. Associate with $E \in \mathfrak{W}_{\text{sl}}$ the class

$$D_E = \{\xi \in \text{Ord} : p_\xi <_{\text{lex}} E\}.$$

If $E <_{\text{lex}} F$, then $D_E \subseteq D_F$, and the inclusion is strict: Corollary 18.7 supplies a code strictly between E and F . At the least ordinal belonging to $D_F \setminus D_E$, the characteristic functions therefore have values 0 and 1, respectively. This proves the second order embedding. Both constructions use elementary class comprehension with the input word or enumeration as a class parameter. $\square$

Bi-embeddability must not be replaced by isomorphism. Binary class words have a least and a greatest word and have adjacent pairs: after a set prefix, the word with next bit 0 and all subsequent bits 1 immediately precedes the word with next bit 1 and all subsequent bits 0. By contrast, $\mathfrak{W}_{\text{sl}}$ is dense and has no endpoints.

[merge] *Second route (source 12)*. Source 12's Section 7 embeds $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ into binary class words by concatenating self-delimiting codes of the entries, instead of source 11's characteristic function of a dense class; the other embedding is source 11's pair swap.

18.6 Source 12's route to the binary-class representation

Let $\mathbb{B}_{\text{Ord}}$ denote the predicate of class subsets of Ord , regarded as binary functions on Ord and compared at their least disagreement. This is again notation for a predicate and relation on classes, not a powerclass object.

Lemma 18.16 (A self-delimiting numerical code). *Replace every minus in a sign sequence by 00, every plus by 11, and append the terminal block 01. The resulting binary ordinal word $c(s)$ has length $2 \cdot \text{bd}(s) + 2$. Distinct codewords are prefix-incomparable, and their first-difference comparison agrees with numerical comparison of the surreals.*

Proof. The three two-bit blocks have order $00 < 01 < 11$, exactly matching $- < \text{end} < +$. At a sign disagreement the first different block decides. When one sign sequence ends, its terminal block differs from the next sign block of the other. Hence neither complete codeword can be a prefix of another, and the numerical comparison is preserved. $\square$

Theorem 18.17 (Binary-class equimorphism). *In the sense of uniform class transformations,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}) \simeq_e \mathbb{B}_{\text{Ord}}.$$

Proof. For $E : \text{Ord} \rightarrow \mathbf{No}$, concatenate $c(E(\alpha))$ over all ordinals. The cumulative length before each stage is a set ordinal. All blocks are nonempty, so the cumulative lengths are unbounded in Ord and the result is a binary class on all of Ord . If E, F first disagree at α , their earlier block lengths and bits agree, and the code of that entry decides the same inequality by lemma 18.16.

For the converse, use the disjoint pairs $a_\alpha < b_\alpha$ in the proof of theorem 18.20. For a binary class A , list the pair in ascending order if its bit is 0, and descending order if its bit is 1. Put pair blocks in ordinal order. The first different bit gives the first different pair orientation and the correct lexical inequality. All labels occur exactly once. $\square$

This is not an isomorphism assertion. The binary-class order has least and greatest elements and adjacent pairs, such as a zero followed by all ones versus a one followed by all zeros. By note 18.6, $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ has no endpoints or adjacent pairs. The distinction between equimorphism and isomorphism is already visible at this level.

Corollary 18.18. *The higher-order order of all class initial segments of $\mathbf{No}$, ordered by inclusion, embeds into $\mathfrak{W}_{sl}$. In particular, this holds for any specified convention of Dedekind cuts. Conversely, each $E \in \mathfrak{W}_{sl}$ determines the initial segment $\{p \in \mathcal{P}_{bd} : p <_{lex} E\}$, and these initial segments separate distinct enumerations.*

Proof. For a class initial segment $C \subseteq \mathbf{No}$, form $e_0^{-1}[C] \subseteq \text{Ord}$ and apply the pair-swap embedding of Theorem 18.15. Proper inclusion of initial segments gives proper inclusion of their inverse images, which implies the correct lex comparison of their characteristic functions. The converse is the dense-class construction in the second half of that theorem. $\square$

This corollary does not form a GB class of all Dedekind cuts of $\mathbf{No}$. Each individual cut is an allowed class parameter; the rule associates an enumeration to it and compares any two such results. An encompassing object belongs to a higher type or to an explicitly larger ambient universe. Nor does the corollary assert that the natural inclusion of $\mathcal{P}_{bd}$ in $\mathfrak{W}_{sl}$ is a Dedekind completion: an order embedding of cuts need not preserve their intended suprema or their original positions relative to the embedded line.

There is a useful description of the natural cuts produced by an enumeration outside the bounded-support family. If $E \in \mathfrak{W}_{sl}$ has no code in $\mathcal{P}_{bd}$, then

$$L_E = \{p \in \mathcal{P}_{bd} : p <_{lex} E\}, \quad U_E = \{p \in \mathcal{P}_{bd} : E <_{lex} p\}$$

partition $\mathcal{P}_{bd}$ into two nonempty classes. Proposition 18.12 shows that L_E has no set-sized cofinal subset and U_E has no set-sized coinital subset, while ordinal-indexed class families suffice on both sides. This cut is unfilled in $\mathcal{P}_{bd}$: a putative filling code and E would be separated by another bounded-support code. Transporting the cut through an isomorphism $\mathcal{P}_{bd} \cong \mathbf{No}$ therefore exhibits a surreal class cut with neither side controlled by a cofinal or coinital set. This is a description of individual additional enumerations relative to the fixed dense class, rather than a classification of all surreal class cuts.

Theorem 18.19 (No exhaustive uniform class of codes). *There is no ordinary class of set codes with a single class evaluation relation whose evaluated rows include every member of $\mathfrak{W}_{sl}$.*

Proof. If such a coding existed, global choice would enumerate its codes by ordinals, with repetitions if the class of codes were a set. Write the resulting row relation as $\text{Eval}(\xi, \alpha, x)$. Use the sorted pairs $u_\xi < v_\xi$ from Theorem 18.15, and define the class

$$D = \{\xi : \text{Eval}(\xi, 2 \cdot \xi, u_\xi)\}.$$

The associated pair-swap enumeration T_D differs from row ξ on the pair $(2 \cdot \xi, u_\xi)$. If that pair belongs to the old row, it does not belong to the graph of T_D , whose value there is v_ξ . If it does not belong to the old row, it belongs to the new graph, whose value there is u_ξ . Thus T_D is a member of $\mathfrak{W}_{sl}$ omitted by the proposed exhaustive list. The argument even permits invalid or nonfunctional rows in the evaluator. $\square$

The theorem explains why class universality of $\mathbf{No}$ cannot simply be applied to the full collection $\mathfrak{W}_{sl}$. It also pinpoints the limitation of coding only bounded modifications, or any other one uniformly evaluable family of set codes: such a family may be order-dense and may realize all set-indexed cuts, while still omitting global enumerations by diagonalization.

[merge] *Strengthening (source 12).* Source 12's Section 5 proves that no uniform family of class well-orders indexed by a class of sets contains every class well-order of $\mathbf{No}$, and that the missing one can be chosen set-like; its corollary 18.21 is theorem 18.19. Its subsection on definable classes explains why enumerating formulas does not evade the theorem.

18.7 Source 12’s strengthening: why the collection of all global well-orders is not a class

The formal obstacle is not that individual class relations are illegal. They are legitimate class variables. The obstacle is attempting to make them the elements of an ordinary class of the same theory.

A useful legal replacement is a *uniform family*. For example, $F \subseteq I \times \text{Ord} \times \mathbf{No}$ is a class of triples of sets, and its section F_i can be a class bijection for every $i \in I$. Similarly a class of triples (i, x, y) can code a family of class order relations. The index class I may be proper. These are single classes, not functions whose values are class objects.

Theorem 18.20 (No exhaustive class-coded family). *In GBC, no uniform family of class well-orders of $\mathbf{No}$ indexed by a class of sets contains every class well-order of $\mathbf{No}$. In fact, for every such family there is a set-like class well-order not among its sections.*

Proof. Global choice permits the family to be reindexed by an ordinal or by Ord . Extend an ordinal-indexed family to an Ord -indexed one by repeating an arbitrary default well-order; an empty family is trivial. Write the resulting relations as R_α .

Fix a bijection $B : \text{Ord} \rightarrow \mathbf{No}$ and partition its range into the pairs

$$\{B(2 \cdot \alpha), B(2 \cdot \alpha + 1)\} = \{a_\alpha, b_\alpha\}, \quad a_\alpha < b_\alpha.$$

Here multiplication is ordinal multiplication. The blocks of length two partition Ord . Define a new enumeration E by placing the two labels of pair α in the order *opposite* to their relative order in R_α . Keep the pair blocks in increasing α order.

This is a class bijection $\text{Ord} \rightarrow \mathbf{No}$, defined elementarily from the family and B . Its induced well-order disagrees with R_α on the relative order of a_α, b_α . Thus it is not any section of the family. $\square$

Corollary 18.21 (No uniform set-code solution). *There is no class D of set codes together with a uniform class decoder whose decoded relations are exactly all class well-orders of $\mathbf{No}$.*

This diagonal theorem applies equally to an attempted exhaustive coding of all class subsets of Ord through the pair construction. It does not say that an externally specified set model has too many classes to form an external set. It says that the corresponding exhaustive decoder is not available *internally at the same size level*.

18.7.1 Definable classes and the truth-predicate trap

Formula codes and set parameters are sets, but the semantic map “formula plus parameters $\mapsto$ its extension in the universe” is not a uniformly available truth predicate for the universe. For each fixed formula one may define a class. One may also work externally with all definable classes of a chosen model. Neither maneuver supplies an internal universal decoder defeating [theorem 18.20](#). The same caution applies to quantifying over all possible Lean universe levels: universe polymorphism is not an internal type of all universes.

18.8 Omitted ranges and the transition to raw well-orders

Theorem 18.22 (An omitted-range cut). *Fix $r \in \mathbf{No}$ and a class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$. There are increasing and decreasing ordinal-indexed families (a_α) and (b_α) in $\mathcal{P}_{\text{bd}}$ such that*

$$a_\alpha <_{\text{lex}} b_\beta \quad (\alpha, \beta \in \text{Ord}),$$

but no member of $\mathfrak{W}_{\text{sl}}$ lies strictly between both entire families. The raw well-order that lists F and then r has type $\text{Ord} + 1$ and is the unique member of $\mathfrak{W}_{\text{all}}$ between these families.

Proof. Use the approximants from Proposition 18.12. They satisfy $a_\alpha <_{\text{lex}} F <_{\text{lex}} b_\beta$ for every pair of indices. If a global bijection E lay between both families and differed from F first at δ , an approximant with index greater than δ would place E on the wrong side. Therefore E would equal F at every ordinal, which contradicts its surjectivity.

Let R be the raw order obtained by appending r after the enumeration F . Its set-like initial part is precisely F : every point listed by F has set-sized predecessors, whereas r has a proper class of predecessors. Its comparisons with the approximants are decided at their prescribed ordinal first differences, so it lies between both families. Conversely, the same first-difference argument forces the core F_S of any raw order S between the families to equal F . By the core decomposition of Theorem 15.1, its residual order must be the unique order on the singleton $\{r\}$. Thus $S = R$. $\square$

The lower and upper families here are indexed by all ordinals, not by sets. Accordingly, the missing interpolant does not contradict Theorem 18.5. It gives a concrete boundary between the set-cut saturation of the set-like comparison and the additional points provided by non-set-like global well-orders.

Example 18.23 (Two omitted points produce adjacent raw orders). Let $r < s$ and choose $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r, s\}$ bijectively. The approximants to F have exactly two raw interpolants: append r, s in that order, or append s, r . Both raw orders have type $\text{Ord} + 2$, and the first is lexicographically smaller. They are adjacent in $\mathfrak{W}_{\text{all}}$. Indeed, a raw order between them cannot have a different core, since an ordinal first difference would put it on the same side of both. With core F fixed, its residual order is a well-order of a two-element set, and these are exactly the two displayed possibilities. Thus the density theorem for $\mathfrak{W}_{\text{sl}}$ does not extend to all raw well-orders.

Corollary 18.24 (Exactly which raw set cuts admit bounded completions). *Let $(A_i)_{i \in I} <_{\text{lex}} (B_j)_{j \in J}$ be a uniformly set-indexed strict cut in $\mathfrak{W}_{\text{all}}$. A member of $\mathcal{P}_{\text{bd}}$ lies between the two sides if and only if*

$$F_{A_i} \neq F_{B_j} \quad (i \in I, j \in J),$$

where F_R is the canonical ordinal enumeration of the set-like initial part of a raw well-order R . The condition is vacuous when a side is empty.

Proof. If the cross cores are different, their comparisons are decided at ordinal first differences and have the same direction as the given raw comparisons. Apply Theorem 18.5 to these uniform families in $\mathfrak{J}$. Every comparison of the resulting completion with a raw input is already decided within its core, giving the required raw inequalities.

Conversely, suppose a cross pair shares a core F . This core cannot be surjective: a surjective core determines the entire raw order, contradicting the strict comparison of the pair. Any $E \in \mathfrak{W}_{\text{sl}}$ therefore differs from F at an ordinal coordinate. That first difference places E on the same side of both raw orders, so E cannot lie between them. This excludes a bounded-support completion in particular. $\square$

The criterion isolates the obstruction without appealing merely to size. Distinct ordinal cores can be separated by one bounded change of the baseline, even for a whole set-indexed cut. Two opposite sides with the same core require information in the residual non-set-like order, beyond every ordinal coordinate available to a global set-like enumeration.

Part V

Foundations, models and formalization

19 Formalizing the subject without changing its meaning

19.1 A bounded development in ordinary set theory

For any ordinal θ , use the set $X = \mathbf{No}_{<\theta}$. The well-order relations on X form a subset of $\mathcal{P}(X \times X)$. Their lexicographic comparison can be defined either by unique ordinal enumeration or by the predecessor formula. This is ordinary ZFC mathematics. No inaccessible cardinal, class theory, or new logical sort is required.

This approach is particularly good for proving the general alphabet theorem, investigating cardinal spectra, and checking how a proposed construction changes birthdays. One should distinguish $\mathcal{W}(X)$, which permits every possible ordinal type of cardinality $|X|$, from its minimal-length stratum. Their cardinalities agree, but their jumps and finer order structure need not.

A family of results for every θ does not automatically give a coherent global map. To take a union, the maps must agree on overlaps. The repository's bounded-surreal nonembedding example makes this restriction concrete.

19.2 Second-order class theory

In GB or GBC, the formula

$$\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee \text{Lex}(R, S) \vee \text{Lex}(S, R)$$

is meaningful with class variables. One can prove trichotomy, transitivity, existence theorems, and uniform-family embedding theorems in this language. It is unnecessary to form a class whose members are R, S .

When a family is indexed by a class I of sets, one class relation $T \subseteq I \times \mathbf{No} \times \mathbf{No}$ supplies its rows. If the rows are pairwise distinct, substituting them into the elementary predecessor formula produces a class linear order on I . With repeated rows, one instead has the corresponding comparison on codes modulo equal evaluations. This is the correct format for most class-sized examples and applications.

In contrast, an expression such as

$$\{R : R \text{ is a class well-order of } \mathbf{No}\}$$

does not define an ordinary GB class of members. Our fraktur notation abbreviates a domain of class quantification. The diagonal theorem shows why a single uniform set-code replacement cannot recover the full domain either.

19.3 Why KM does not erase the extra level

KM strengthens comprehension by allowing class quantifiers, but its classes still have *sets* as members. It therefore does not turn the totality of class well-orders into an ordinary class. A genuine object containing arbitrary classes as members requires another sort, or a containing universe in which those classes become sets.

This distinction matters when one wants an arbitrary topology of higher-order open collections, a measure on all global well-orders, or a power object of their totality. Such notions require an explicit

ambient theory. Hyperclass forcing provides one example of higher-level foundational practice [28]; it is not a reason to silently ignore the level distinction in bare KM.

There is also a separate recursion issue. Extracting the canonical initial Ord-part of one raw well-order is an elementary class construction. Iteratively deleting proper-class tails through an arbitrary transfinite sequence would have class-valued stages and needs its own analysis. General comparability of abstract class well-orders by initial-segment isomorphisms is another stronger topic; ETR implies it [27]. None of that machinery is needed for the labeled comparison proved in this article.

19.4 An explicit containing-universe model

A useful concrete model assumes a strongly inaccessible cardinal κ in an ambient universe and takes

$$U = V_\kappa, \quad \mathcal{C} = \mathcal{P}(V_\kappa).$$

This is a sufficient setting for interpreting a full class universe. The internal surreal class becomes the ambient set

$$\mathbf{No}^U = \mathbf{No}_{<\kappa}, \quad |\mathbf{No}^U| = \kappa.$$

All its ambient well-orders form a set of cardinality 2^κ . Those that are internally set-like have ambient order type κ ; unrestricted well-orders may have any ambient ordinal type of cardinality κ .

To see the set-like assertion, every initial segment of a κ -enumeration has fewer than κ members, with ranks bounded below κ ; regularity puts it in U . Conversely, an ordinal type strictly exceeding κ has an initial segment of cardinality κ , which cannot be a set of U . A smaller type cannot enumerate the entire carrier.

The bounded-support class becomes an ambient order of cardinality κ , while the full κ -permutation order has cardinality 2^κ . Thus a κ -sized dense suborder can coexist with a larger order of all enumerations. The exact ambient ordinal cutoff remains κ^+ .

Every ordinal of U is below κ . There is no contradiction between internally embedding all ordinals and having an ambient successor-cardinal barrier. Inaccessibility is an additional assumption for this full universe example; it is not required for an individual birthday fragment.

19.5 A strict nonembedding result

The containing-universe interpretation makes a further comparison decidable by ordinary set-order invariants.

Theorem 19.1 (Nonembedding in a full universe). *In the preceding full interpretation $U = V_\kappa$, $\mathcal{C} = \mathcal{P}(V_\kappa)$, with ambient ZFC and κ strongly inaccessible,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}^U) \hookrightarrow \mathfrak{W}_{\text{all}}(\mathbf{No}^U), \quad \mathfrak{W}_{\text{all}}(\mathbf{No}^U) \not\hookrightarrow \mathfrak{W}_{\text{sl}}(\mathbf{No}^U).$$

These are assertions about the actual ambient sets of internal class well-orders and arbitrary ambient order embeddings.

Proof. The first embedding is inclusion. Put $X = \mathbf{No}^U$. The target of the proposed reverse embedding is the dense order of all bijections $\kappa \rightarrow X$. Its bounded-support suborder $D = \mathcal{P}_{\text{bd}}^U$ is dense and has ambient cardinality κ . For the upper bound, codes with domain $\lambda < \kappa$ are subsets of $\lambda \times \lambda$; strong inaccessibility bounds their number below κ , and there are κ possible domains. Transpositions give the matching lower bound.

Fix labels $a < b$ in X . There are 2^κ bijections $F : \kappa \rightarrow X \setminus \{a, b\}$, by the fixed-alphabet cardinality theorem. Each determines a convex adjacent pair

$$R_F = F + (a, b) <_{\text{lex}} S_F = F + (b, a)$$

in the unrestricted order. Distinct F 's determine distinct, disjoint convex fibers; one entire pair precedes the other.

If $\Phi : \mathfrak{W}_{\text{all}}(X) \hookrightarrow \mathfrak{W}_{\text{sl}}(X)$ existed, the intervals

$$(\Phi(R_F), \Phi(S_F))$$

would be 2^κ pairwise disjoint nonempty open intervals in the target. Each contains a point of D . Choosing the least such point under a fixed well-order of D would inject 2^κ into κ , a contradiction. $\square$

Thus equal cardinality 2^κ and equal ordinal embedding cutoff κ^+ do not imply equimorphism. The useful invariant here is the size of a family of disjoint open intervals relative to a dense suborder. The repository already uses related separability and cellularity arguments for the real alphabet [1, Remark `lwo:sp:rem:conditional`]; the argument above supplies the precise surreal-universe version.

There is also a direct class-theoretic obstruction, provided the inverse evaluation relation can be formed by impredicative comprehension.

Theorem 19.2 (No definable uniform embedding in $\text{KM} + \text{GC}$). *For every fixed formula, with fixed set and class parameters, that defines a total transformation $R \mapsto \Phi(R)$ from $\mathfrak{W}_{\text{all}}(\mathbf{No})$ to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, the theory $\text{KM} + \text{GC}$ proves that this transformation is not an order embedding. The transformation is specified relationally: the defining formula assigns a unique output class graph to each input class well-order. This is a schema for fixed formulas, not quantification over a third sort of arbitrary transformations.*

Proof. Suppose such an order embedding were defined. Fix $a < b$ and a global well-order of the set codes in $\mathcal{P}_{\text{bd}}$. For each class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, b\}$, form the adjacent raw pair R_F, S_F from the preceding proof. Density of $\mathcal{P}_{\text{bd}}$ in $\mathfrak{W}_{\text{sl}}$ supplies codes strictly between $\Phi(R_F)$ and $\Phi(S_F)$. Let q_F be the globally least such code. Comprehension in KM makes this a legitimate least-element definition, even if the formula specifying Φ contains class quantifiers. Disjointness of the image intervals implies

$$q_F = q_G \implies F = G.$$

Now use KM comprehension to form the ordinary class relation

$$\text{Eval}(q, \alpha, x) \iff \exists \text{ class } F [F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{a, b\} \wedge q = q_F \wedge F(\alpha) = x].$$

Its rows are empty or the graphs of unique such bijections, and every such bijection occurs. Compose all rows with one fixed class bijection $\mathbf{No} \setminus \{a, b\} \rightarrow \mathbf{No}$. The resulting single class evaluator, with the ordinary set codes $q \in \mathcal{P}_{\text{bd}}$, contains every global enumeration of $\mathbf{No}$. This contradicts Theorem 18.19. $\square$

The inverse evaluator is the decisive extra comprehension step: its definition quantifies over an arbitrary class F . The preceding proof does not claim that elementary GB comprehension suffices. Nor does it exclude arbitrary external maps between the available-class collections of a nonfull model. In particular, for an externally countable model these collections can both be countable, and its externally dense set-like order can admit an external copy of its unrestricted order. Such a map need not be representable by any uniform definition in the model. This is why the stated interpretation of an embedding matters.

19.6 Internal and external well-foundedness

For a model $(M, \mathcal{A})$, an internally well-founded class relation need not be externally well-founded. An external descending sequence might not be represented by a set in M . Thus comparing the internal theory with an external collection of all relations requires care.

Over GBC, well-foundedness of a class relation can be characterized by the absence of a set-coded descending ω -sequence [27, Section 3]. Keeping the sets and an existing relation fixed while changing only the available classes does not arbitrarily change that relation's internal well-foundedness in suitable models. What can change is which class relations are available at all. External correctness can be ensured by an appropriate well-founded full universe interpretation, as above.

19.7 A practical Lean architecture

The repository already has a universe-indexed sign carrier [7]. Its type structure can be displayed schematically as:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This is mathematical pseudocode using ASCII logical notation, not a new executable Lean declaration. The key point is the level separation: the full carrier is one universe above the types indexing its small sign lengths.

The following existing declarations were inspected:

Module or declaration	Role in the proposed development
<code>SignSequence.ordinalOrderEmbedding</code>	The all-plus ordinal embedding
<code>birthdays_bounded, small_bounded</code>	Control of small parameter families and bounded fragments
<code>SignSequence.not_small</code>	The carrier cannot be treated as small at its ordinal level
<code>SignSequenceCutOperation.cut</code> and <code>cut_realizes</code>	Actual separators for small cut data
<code>SizeObstructions.noUnrestrictedCuts</code>	The obstruction to unrestricted cut filling

These are source-level observations, not results of a fresh build. The current repository contains later arithmetic modules as well; historical proposal notes should not be mistaken for the present implementation status.

A sensible formalization proceeds in four parts.

1. **Set carriers.** For $X : \text{Type } u$, represent well-orders as relations on X together with their laws. Prove the predecessor-based comparison and its equivalence with ordinal enumeration. This avoids transporting across ordinal representatives at every comparison.
2. **Ordinal words.** Define words with small ordinal domains, an injective subtype, ordinal concatenation, and the sign code. Formalize prefix-freeness and first disagreement before proving the cardinal spectrum.
3. **Relative set-likeness.** If X lives in $\text{Type } (u+1)$, express the condition that each predecessor subtype is `Small.{u}`. A well-order of this larger X has its unrestricted order type at the larger ordinal universe. A same-level enumeration must not be inferred without the relative smallness hypothesis.

4. **Uniform families.** Keep an explicit index type and an evaluation relation. The bounded-support codes are sets of ordinal pairs at the selected level. Formulate the interpolation theorem only for small index families and the no-code theorem at its correct larger level.

Lean universes and Mathlib ordinals provide precisely the needed separation [29, 30]. Universe polymorphism is a schema of well-typed constructions, not a term denoting the union of all universes. This article supplies no new Lean file and makes no machine-verification claim.

[merge] Source 12's Section 11 identifies both global orders in the full-class model $(V_\kappa, \mathcal{P}(V_\kappa))$ of [theorem 19.1](#). Its clause (iv) gives a second proof of that theorem's non-embedding: the full order $\mathcal{W}(\mathbf{No}_{<\kappa})$ has binary coding length κ^+ ([theorem 6.8](#)), whereas the set-like order $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ has coding length κ ([theorem 8.3](#), regular case), and coding length cannot decrease under embeddings ([lemma 6.4](#)). Source 11's proof counts 2^κ disjoint intervals against a dense suborder of size κ .

20 A full-class model that connects the two size levels (source 12)

Assume in an ambient ZFC universe that κ is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)),$$

with all external subclasses of V_κ available as its classes. This is a familiar full-class interpretation of strong class set theory: regularity and the strong-limit property ensure the required closure of set operations and class Replacement. It satisfies KM and global choice. We use this as an explicit relative model, not as a claim that inaccessible cardinals are necessary for the earlier GBC theorems.

Theorem 20.1 (Exact external identification). *For the full-class interpretation above:*

- (i) *the internal surreal carrier is externally $\mathbf{No}_{<\kappa}$, and its internal ordinals are the external ordinals below κ ;*
- (ii) *the internal set-like global surreal well-orders, viewed as an external set order, are exactly $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$;*
- (iii) *all internal global surreal well-orders, viewed externally, form exactly $\mathcal{W}(\mathbf{No}_{<\kappa})$;*
- (iv) *both external orders have cardinality 2^κ and ordinal spectrum $\beta < \kappa^+$, but their binary coding ranks are respectively κ and κ^+ . Therefore the full order does not embed into the set-like order.*

Proof. A sign sequence belongs to V_κ precisely when its ordinal domain has length less than κ . Inaccessibility gives $|\mathbf{No}_{<\kappa}| = \kappa$.

Every external relation on $\mathbf{No}_{<\kappa}$ is a permitted internal class: each ordered pair of its labels is an element of V_κ . External well-foundedness implies internal well-foundedness. Conversely, if a relation is externally ill-founded, ambient choice supplies a descending ω -sequence. Its countably many labels have ranks bounded below κ by regularity, so the sequence is an element of V_κ and witnesses internal ill-foundedness. Thus the well-orders agree.

An internal set has external cardinality less than κ . Conversely, a subset of $\mathbf{No}_{<\kappa}$ of cardinality less than κ has bounded ranks and is an element of V_κ . Therefore internal set-likeness means that every proper predecessor segment has cardinality less than κ . An external well-order of κ labels has that property exactly when its order type is κ , proving (ii). Arbitrary external well-orders on those labels have types exactly the ordinals of cardinality κ , proving (iii). Comparisons agree by [theorem 14.7](#). Finally apply [theorems 6.8](#) and [8.3](#) and the monotonicity of coding rank under embeddings. $\square$

20.1 Why this does not contradict surreal universality

Internally, each class linear order has elements that are sets of V_κ and hence externally at most κ elements. The external collection of all class well-orders has 2^κ elements and is not an internal class carrier. Thus it is outside the scope of the internal class-universality theorem.

If $\lambda > \kappa$ is another inaccessible cardinal, the entire external set $\mathcal{W}(\mathbf{No}_{<\kappa})$ can be studied as an ordinary set in V_λ . It then embeds into the larger surreal carrier there by set-order universality. Its labels have moved up one foundational level; no same-level universal object has been created.

The singular cutoff $\beth_\omega$ in [section 8](#) must not be used as though $V_{\beth_\omega}$ were this model. That rank lacks the regularity used in Replacement and in the well-foundedness argument. The singular theorem is an ordinary ZFC theorem about a bounded set, not a full-class model construction at a singular height.

20.2 Henkin classes, countable models, and relativity

If the class collection is some $\mathcal{X} \subsetneq \mathcal{P}(M)$, only those well-orders whose relations lie in $\mathcal{X}$ are admitted. External cardinality and saturation can then differ from the full-class calculation. A countable model may have an externally countable collection of admitted classes and external countable cofinal families, while internally having no *coded set-indexed* cofinal family. The missing external families are simply not internal objects.

For nonstandard or non- β class models, one must additionally separate internal and external well-foundedness. Our exact identification in [theorem 20.1](#) used transitivity, full classes, and regularity; it was not asserted for arbitrary models. The general subtleties of class well-foundedness are discussed in [\[27, §3\]](#).

[merge] Clause (ii) of [theorem 20.1](#) is source 11’s statement that the internally set-like well-orders have ambient type κ , and clause (iii) its statement that unrestricted ones have every ambient type of cardinality κ ([section 19](#)). Source 12’s remark that the singular cutoff $\beth_\omega$ must not be used as though $V_{\beth_\omega}$ were such a model is the counterpart of source 11’s remark that inaccessibility is needed for the full universe example but not for an individual birthday fragment.

[merge] The next section is source 12’s formalization proposal. The modules it names are proposals and do not exist; the existing declarations it cites exist at the current revision, and none states a result of this report ([section 1.5](#)).

21 A formalization architecture with explicit size boundaries (source 12)

21.1 Start with a universe-polymorphic set theorem

The most efficient verified starting point is independent of surreal arithmetic. For a type X with a linear order, represent a well-order on X by its binary relation and a proof of well-foundedness and linearity. Define the common prefix by equality of predecessor sets and equality of the restricted relations, exactly as in [section 14.1](#). Prove maximality, distinct next labels, and transitivity of comparison. This gives a generic theorem reusable for reals, surreal cutoffs, and other ordered alphabets.

For a fixed ordinal domain, represent an enumeration as an equivalence between an ordinal’s carrier and X . For all well-orders of a small X , using relations on X often avoids unnecessarily quantifying over every ordinal in a larger universe. The ordinal enumeration can be recovered later

from the well-order type. This is a representation choice, not an identification of all ordinals with a small type.

21.2 Use the existing sign-sequence foundations selectively

The inspected repository file

```
Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean
```

contains, among others,

```
isPrefix_of_minimum_birthday
existsUnique_prefix_of_ordConnected
existsUnique_minimum_birthday_of_ordConnected
existsUnique_simplest_separator
```

[10]. Its comments explicitly distinguish uniqueness of a simplest separator from existence of separators for arbitrary small cuts. The last theorem assumes a separator exists. It must not be imported as though it independently established all cut-filling instances.

A suitable interface for the order-theoretic development needs a birthday map, a numerical linear order, sign concatenation and truncation, the first-difference comparison, and small-cut existence with a birthday bound. Field operations can be omitted except in illustrative examples.

21.3 Suggested proof modules

Module	Principal obligations
CommonPrefix	The elementary definition; initiality; maximality; next-label inequality; generic full lexical linearity.
ExhaustiveLex	Prefix-freeness; equivalence with ordinal first-difference comparison; transport between relations and enumerations.
BinaryCubes	Cylinder fusion; strict hierarchy; characteristic-cut embeddings; precise ordinal concatenation.
PermutationCoding	Pair orientation, product scheduler, transpositions, and exhaustive output.
BirthdayCutoffs	Size of $\mathbf{No}_{<\kappa}$; prefix-free numerical code; cone partition; regular and singular sums.
ClassFamilies	A formal set/class model, sections of class relations, baseline enumeration, and diagonal non-enumerability.
UniverseModels	A chosen inaccessible-rank interpretation; internal/external well-foundedness and identification of the two orders.

The finite common-prefix theorem should be proved directly for relations, not by first developing all of ordinal arithmetic. The scheduler proof should record both surjectivity and the first-row-appearance invariant; a bare cardinal pairing equivalence is insufficient. The singular theorem must use ordinal sums, never a cardinal cast that silently loses order.

21.4 Universe accounting

Schematically, a sign-sequence type built from small ordinal lengths has one universe level for its ordinal index and potentially a higher level for the total carrier. One should retain explicit levels rather than promise a concrete declaration before checking the chosen ordinal API. A bounded carrier with a specified small birthday bound is the cleanest first implementation.

Three operations must remain visibly different: taking a subtype of one fixed surreal carrier; lifting an existing carrier to a larger universe; and constructing all surreals of the larger universe. The third adds new birthdays and is not merely a type lift. Likewise, a type of relations on a fixed carrier can exist at a higher level without becoming a same-level class of all class relations.

For global theorems, two safe routes are available. One may formalize GB or GBC models and express theorems about their class predicates. Or one may formalize the bounded analogue inside a hierarchy of explicit universes and separately prove its interpretation in a set/class model. Neither route requires an inconsistent type of all types.

21.5 Axioms, kernels, and honest validation

A full formalization should record the uses of classical logic, set choice, global choice, and any ambient model-existence assumptions. The common-prefix comparison should not acquire an ETR assumption merely because abstract class order comparison sometimes uses one. The inaccessible-rank example is conditional on that ambient cardinal, whereas the bounded singular theorem needs no inaccessible cardinal.

The present package includes exact executable tests of finite shadows: sign coding, pair coding, the scheduler, the common-prefix formula, and finite adjacency. These are useful implementation checks and counterexample detectors. They are not kernel-checked proofs of any infinite or class theorem, and no supplied file is advertised as such.

[merge] The four plans agree on a first milestone that is independent of surreal arithmetic: the predecessor or common-prefix comparison of well-orders on a small carrier, prefix-freeness and the successor-cardinal obstruction, then bounded-support prefix completion and small-family interpolation, reusing the existing small-cut interface. This is a proposal recorded for the collection's formalization work; no such module exists.

Questions, conclusions and status

22 Research questions

[merge] The sources' questions are printed per source in the following subsections, as proposed continuations, not as a census of open problems. Source 12's question on the weak-choice strength (remark 22.15) is answered by theorem 13.1 and printed as an answered remark. Source 11's research question 22.7 and source 12's research question 22.13 ask related questions on invariants at cutoffs; source 11's research question 22.2 and source 12's research question 22.16 on minimal axioms; source 11's research question 22.3 and source 12's research question 22.14 on proper-class blocks; source 11's research question 22.4 and source 12's research question 22.17 on automorphisms; and source 11's research question 22.8 and source 12's research question 22.18 on the class realization. Source 12's research question 22.11 continues the reals report's question "Other ground orders", which theorem 8.3 answers in part.

22.1 Source 11's research questions and topics

The questions below are proposed extensions of the present development. Except where a source is explicitly named, they are not asserted to be established open problems in the literature. Several ask for a stronger classification than the proved results supply.

22.1.1 A precise description of the represented cuts

Fix the dense ordinary class $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$. Every set-like global enumeration determines a lower cut of $\mathcal{P}_{\text{bd}}$, and so does every injective nonexhaustive Ord-word. The omitted-range example shows that the first collection does not fill every cut generated by the second.

Research question 22.1. Characterize, using the prefix tree and its numerical labels, exactly which class cuts of $\mathcal{P}_{\text{bd}}$ arise from injective Ord-words, and which of these words are exhaustive.

A useful first target is a necessary-and-sufficient compatibility condition on the intersections of a cut with prefix cylinders. The challenge is to separate missing labels, impossible repeated labels, and cuts that escape every coherent branch. This should first be stated for a fixed uniform class cut, before quantifying over an entire hyperclass of cuts.

22.1.2 The embedding relationship between the two global orders

The set-like fragment and binary class words are equimorphic, while the unrestricted order has additional convex fibers. Theorem 19.1 rules out a reverse embedding in the full universe interpretation. Theorem 19.2 also rules out a definable uniform reverse embedding in $\mathbf{KM} + \mathbf{GC}$.

Research question 22.2. What is the weakest comprehension scheme over GBC that proves the uniform nonembedding result? Can the inverse-evaluation step be replaced by an elementary argument, or is a genuine strengthening necessary?

A precise project should specify the syntactic complexity of the proposed transformation and of the comprehension used to invert its code assignment. In the bounded setting one can instead compare the density and cellularity of all well-orders of $\mathbf{No}_{<\kappa}$ with their minimum-length stratum, especially when κ is not inaccessible. The full-universe case is already settled above; the remaining questions concern strength and wider hypotheses.

22.1.3 Iterated residual decomposition

One extraction of the canonical initial Ord-part requires only elementary class comprehension. A residual proper class has another such part, suggesting a hierarchy of successive blocks.

Research question 22.3. What is the exact recursion strength needed to organize iterated residual decompositions uniformly, and what class order types can occur as their block-index orders?

An initial project could analyze finitely many blocks and specified set-length constructions, recording where the recursion state becomes a class. The general case should be compared with known ETR results on class well-order comparability, rather than assumed to follow from the one-step theorem.

22.1.4 Recovering the prefix structure from the bare order

The definitions exhibit a labeled tree of prefixes. An automorphism of the bare lexicographic order need not obviously preserve its displayed labels or cylinders.

Research question 22.4. Which prefix cones, canonical-core fibers, and finite permutation blocks are definable from the linear order alone? What extra predicates make the reconstruction unique?

Adjacency directly detects finite stretches, but a singleton fiber can be exhaustive or can omit one label. Thus local block size alone cannot recover set-likeness. An analysis of the automorphism group may clarify what the order remembers about the original enumeration.

22.1.5 The bounded-support choice of baseline

Every baseline enumeration gives an ordinary class $\mathcal{P}_{\text{bd}}$ abstractly isomorphic to **No**. If two baselines differ on a proper class of positions, their bounded-support subfamilies can be disjoint.

Research question 22.5. How do the two dense classes sit inside the same $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, and when does an automorphism of that higher-order order carry one onto the other?

The abstract isomorphism of the two dense orders does not automatically extend to all global enumerations. The represented-cut problem is likely relevant. A restricted first case is a baseline change obtained by an order automorphism of the numerical surreal line.

22.1.6 Sharper birthday bounds and minimal codes

The explicit code gives a usable bound

$$\text{bd}(E(f)) = \sum_{i < \text{dom } f} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

It is designed for a transparent proof, not asserted to be optimal.

Research question 22.6. Which birthday bounds are optimal for uniform embeddings of injective surreal words? Can shorter prefix-free coefficient codes preserve useful simplicity relations?

Regular cutoffs permit same-stage compression; singular cutoffs show why a cardinal bound alone is insufficient. A sharpened result should specify whether it preserves only numerical order, prefix extension, simplicity, or all of these.

22.1.7 Saturation of more general bounded alphabets

If X is an η_κ order of cardinality κ , its minimal-length permutation order inherits η_κ . Birthday fragments often have cardinality $2^{<\kappa}$, which may exceed κ .

Research question 22.7. Determine the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$, where $\lambda = 2^{<\kappa}$, without assuming $2^{<\kappa} = \kappa$.

The issue combines the size of a used prefix with the number and birthdays of possible numerical separators. The real-alphabet formulas in the precursor report cannot settle this case by substitution.

22.1.8 Class extensions and available global orders

Two models may have the same sets and hence the same individual surreal numbers, but different available classes and different global well-orders.

Research question 22.8. Which features of the lexicographic theory are invariant under extensions of the class part, and how does the represented-cut collection change?

Uniform formulas such as first divergence remain the same for the existing relations. Existence of additional enumerations changes the larger totality. This offers a way to distinguish properties of the labeled surreal order from properties of the surrounding class theory.

22.1.9 A verified first milestone for ProveIt

Research question 22.9. Can the explicit word compression and the direct predecessor comparison be formalized first, with a small, reusable interface independent of surreal arithmetic?

These are concrete, finite-dependency targets: ordinal concatenation, numerical first difference, injectivity, and well-order predecessor sets. The next milestone is the general successor-cardinal obstruction, followed by bounded-support completion and the small-family interpolation theorem. This ordering keeps the central algorithms separate from a formalization of an entire higher-order set theory.

22.1.10 Uniform mathematical constructions indexed by well-orders

The Kanovei–Shelah method suggests using a definable linear order of codes to index constructions without selecting one privileged code. The surreal setting raises additional smallness conditions.

Research question 22.10. Which finite-support or bounded-support constructions can be indexed by a uniform class family of global well-orders, and which require a genuine higher-order index object?

A first investigation should specify its support sets, evaluation maps, and quantifier domains before adding algebra or model theory. The diagonal theorem prevents an exhaustive class-code shortcut, but leaves many useful uniform families available.

[merge] Status of source 11’s questions after the merge: all remain open. Source 12’s dichotomy (theorem 8.3) computes the binary coding length of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ in every case, but not the cut spectrum or the point characters that research question 22.7 asks for.

22.2 Source 12's research questions and prioritized program

The following are proposed continuations. Except where an older question is explicitly attributed, they are not claimed to be recognized open problems in the literature.

Research question 22.11 (All fixed enumeration types). Determine $\ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}))$ for every ordinal α of cardinality $2^{<\kappa}$, not only the minimal type. Can the answer be expressed by a precise ordinal-sum invariant of admissible entry-code lengths, together with the finite terminal permutation factor?

The singular theorem identifies the first obstruction that any such formula must capture. A useful initial target is $\alpha = \mu \cdot 2$ and then $\mu \cdot \gamma + n$ for finite n .

Research question 22.12 (Noncardinal birthday bounds). For an arbitrary infinite ordinal θ , classify the minimal permutation order of $\mathbf{No}_{<\theta}$. Which features of the ordinal tail of θ , beyond $|\theta|$ and $|\mathbf{No}_{<\theta}|$, survive in its coding rank and local character?

The cone argument for an initial cardinal relies on closure under $i + 1 + \alpha$ below the cutoff. At a noncardinal cutoff that closure can fail and should be replaced by a stratified family of cones.

Research question 22.13 (Isomorphism, not merely equimorphism). Classify the order topology and isomorphism invariants of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$. At regular cutoffs, and at singular strong limits, what are its density, cellularity, cut-character spectra, and degrees of homogeneity? Which distinguish it from other orders of the same binary rank?

The binary cube itself cannot be the literal answer: it has endpoints and adjacent pairs, while the surreal permutation order has different local behavior. A dense suborder equimorphic with a cube can retain substantial information not measured by rank.

Research question 22.14 (The hierarchy of proper-class blocks). Study the subpredicates of $\mathfrak{W}_{\text{all}}(\mathbf{No})$ with types $\text{Ord} \cdot \gamma + \alpha$ for fixed set ordinals γ, α . Give explicit, size-safe embedding and nonembedding theorems between these subpredicates and the set-like order.

The inaccessible-model calculation provides concrete bounded targets and shows that the full order is strictly more complex than the set-like order there. Extending binary fusion to uniform transformations on class predicates may require stronger comprehension; that dependence should be proved, not assumed away.

Remark 22.15 (Source 12's question "The exact weak-choice strength", answered). *Source 12's text:* Remove set-level AC from the base theory in [theorem 13.2](#). What principle is equivalent to a class well-order of the sign-sequence surreals then? Compare it with coherent well-ordering of power sets of ordinals and with uniform ordinal presentations of transitive closures.

[merge] *Answered (2 October 2026)*. Source 11's [theorem 13.1](#) proves the equivalence over GB with no set-level choice: a class well-order of the sign-sequence surreals is equivalent to global choice. Its proof is precisely a coherent well-ordering of the power sets $V_{\beta+1}$, obtained by set-valued recursion from the given well-order on the sign sequences of each length θ ; see [remark 13.4](#).

This is separate from the question whether surreal universality for class linear orders itself implies global choice, raised by Hamkins [\[21\]](#). The present equivalence concerns a well-order of all surreal labels and does not settle that universality question.

Research question 22.16 (Elementary common-prefix theory). Formalize [theorem 14.7](#) over a precisely chosen weak class theory. How little comprehension and class well-foundedness are needed? Can one characterize the same relation directly by a formula avoiding explicit mention of the common-prefix class?

The definition already expands to elementary quantifiers. A mechanized proof would make the separation from abstract class comparability particularly transparent and yield a reusable theorem for other carriers.

Research question 22.17 (Automorphisms and finite condensation). Determine natural automorphism groups of the full lexicographic order, and of the order after collapsing finite residual blocks. Which operations arise from numerical automorphisms or permutations of the underlying surreal carrier, and which genuinely act on enumeration structure?

Finite-tail blocks are now characterized intrinsically by adjacency. That makes condensation a plausible first invariant, but no complete classification of its quotient is claimed here.

Research question 22.18 (Dependence on the class realization). Fix a transitive set universe M and vary its admissible class collection $\mathcal{X}$. Which external order invariants of the admitted global well-orders detect this change? Can the order reconstruct useful information about available truth predicates or class choice functions?

The diagonal obstruction persists internally, while external cardinality can vary. This makes the distinction between full and Henkin semantics a source of mathematical questions rather than merely a formal warning.

Research question 22.19 (Compatibility between birthday cutoffs). Construct and analyze coherent embeddings from the well-order spaces at smaller birthday cutoffs into those at larger cutoffs. Which choices of reserve labels and tail enumerations are functorial, and which preserve finite condensation, coding ranks, or canonical cylinders?

Naively extending every enumeration by an arbitrary tail can change minimal length, and fixed choices need not be compatible across three cutoffs. A coherence theorem would make universe-lifting arguments much more useful in a proof assistant.

[merge] Status of source 12's questions after the merge: [remark 22.15](#) is answered (it is printed above as a remark). [Research question 22.11](#) continues the reals report's question "Other ground orders" for the surreal cutoffs, whose minimal type [theorem 8.3](#) settles. The others remain open.

23 Source 12's conclusion

[merge] Source 11 has no conclusion section; its summary is [section 24](#).

23.1 Source 12's conclusion

There is a rigorous full lexicographic order of all global surreal well-orders, provided it is understood as a relation on class objects, not as an ordinary class containing those objects. Its comparison needs no universal enumeration of abstract class order types: the largest common *labelled* prefix is elementary and sufficient.

The set-like suborder has a particularly uniform geometry. It fills set-indexed cuts, contains every class linear order even locally, and is equimorphic with binary classes on Ord . The unrestricted order adds new proper-class prefix phenomena, including finite adjacent blocks. The diagonal theorem prevents either construction from being exhausted by a same-level class of set codes.

At set-sized birthday bounds the theory becomes quantitative. The full well-order space of a carrier of cardinality μ has exact ordinal spectrum below μ^+ and binary rank μ^+ . The minimal

surreal slice has rank μ , except at singular strong-limit cutoffs, where it has rank $\kappa \cdot \kappa$. This is an order-theoretic effect of cofinal blocks of code lengths, invisible to cardinal counting alone.

The inaccessible-rank interpretation shows exactly how these facts can coexist: internal classes and the external collection of all such classes occupy different levels. Respecting those levels removes the apparent paradox without weakening the embedding and universality questions into trivialities.

24 Results, assumptions, and verification status

Result	Foundation used here
Fixed-alphabet cardinality and ordinal spectrum	ZFC; existing repository theorem
Explicit set-length word compression	ZF, as a definable class construction
Local word characters and set-cut arguments	Surreal set cuts; ZFC for cardinal-valued characters
Any class well-order of No implies global choice	GB with ZF set part
Raw comparison, canonical initial part, finite-jump criterion	GB, with the given class orders
Bounded-support completion and interpolation	GBC, with a fixed global enumeration
$\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ and class-order universality	GBC, by set-valued back-and-forth
Binary class coding and no exhaustive uniform evaluator	GBC, interpreted with class parameters
Unrestricted order does not embed into its set-like fragment	Full inaccessible universe; or $\text{KM} + \text{GC}$ for definable uniform maps
All higher-order collections treated as actual objects	An explicitly chosen higher-order or containing-universe setting

The proof dependencies are short. Set cuts yield fresh coefficients after deleting a set. Fresh coefficients and bounded-support completion yield strong interpolation. Strong interpolation yields density, local cofinality statements, the **No**-isomorphism, and cut coding. Independently, the predecessor comparison yields convex cones, the canonical initial-part decomposition, and the finite-residual jump theorem. Pair swaps supply both positive binary embeddings and the diagonal obstruction.

The bundled Python script checks finite instances of the two comparison rules and of the sign encoding. It checks 465 coefficient-code comparisons, 79,800 word-code comparisons, and 266,272 pairs of finite permutations, with the predecessor rule checked in both directions and the adjacency criterion compared with the finite lexicographic order. These checks help detect a reversed delimiter or an incorrect finite comparison. They do not verify an ordinal induction, a class comprehension argument, or a cardinal nonembedding theorem.

The article contains full mathematical proofs for the claims it promotes to theorems. It does not claim that the new results have been checked in Lean, that the whole repository has been audited, or that every research question listed here is a previously published open problem.

[merge] The other sources' ledgers follow; each records the hypotheses its source used, not that each is necessary.

24.1 Source 12's proof-dependency and assumption ledger

Result	Stated setting	Essential input
Common labelled prefix and full lexical comparison	GBC; comparison is elementary	Existing well-orders; class minima; ambient linear order.
Global-choice equivalence	GB + set AC	Ordinal coding of transitive closures; restriction of a surreal well-order to a set of codes.
No exhaustive class-coded family	GBC	A baseline Ord-bijection; elementary diagonal pair orientation.
Set-like cut interpolation	GBC	Set-sized discrepancy bound; surreal small-cut filling; prefix extension.
Local class universality	GBC	Set-valued forth and back-and-forth recursions; global enumeration.
Finite-tail adjacency	GBC	Label transpositions; finite residual cylinders; no infinite order well-ordered in both directions.
Set-sized full spectra	ZFC	Pair coding; regularity of μ^+ ; cylinder-fusion nonembedding.
Minimal-slice singular transition	ZFC	Cone partition; scheduler; ordinal block sums; regularity or singular cofinality as appropriate.
Full-class rank model	Ambient ZFC + an inaccessible κ	Closure of V_κ ; all external classes; regularity for descending sequences.

A Source crosswalk and reproducibility

The immutable sources used in the targeted audit are recorded in the accompanying source audit. For convenient verification, the most relevant mappings are:

Source	Relevant content
[1]	Labels <code>lwo:lem:prefix-free</code> , <code>lwo:lem:cylinder</code> , <code>lwo:prop:cardinality</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code> ; bounded sign orders in <code>lwo:sp:thm:surreals</code> .
[3]	Set-coded surreals and bounded birthdays; unrestricted cut obstruction; class-quantification and universe distinctions; set-valued recursion.
[7]	Actual sign carrier, ordinal embedding, smallness obstruction, and bounded fragments.
[9]	Small cut operation, separator properties, and birthday bounds.
[5]	A nearby use of a set-like global well-order and ordinary class-parameter recursion.

To compile the delivered standalone source, run `latexmk -pdf surreal_well_orders.tex` from its directory. All article sections and the bibliography are included in that single file; no separate

bibliography processor is needed. Run `python3 finite_checks.py` for the finite checks. The supplied PDF was compiled from the included sources and inspected after rendering.

[merge] *Delivery names.* The paragraph above describes source 11’s delivered package. Its `surreal_well_orders.tex` is shipped as `article.tex`, which this merged text replaces; its PDF is not shipped (`article.pdf` is a build of this text); its `finite_checks.py` and `finite_checks_results.txt` are `code/11-raw-orders-finite_checks.py` and `data/11-raw-orders-finite_checks_results.txt`; the “accompanying source audit” is `11-raw-orders-repository_audit.md`. Build this report with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex`, and rerun the checks as in [section B](#). The same applies to the reproducibility records of the other sources that follow.

A.1 Source 12: reproducibility and inspection record

The package contains `article.tex`, the compiled `article.pdf`, a build script, an exact finite-check program, its recorded JSON output, and a research-status and source-audit file. The Python program uses only the standard library. The recorded run passed all 180,905 finite comparison cases. Its comparisons are exact; there are no floating-point tolerances or inferred transfinite results. The PDF was compiled and rendered for visual inspection.

The repository audit directly inspected the merged real-well-order README and the sign-sequence simplicity Lean source at the stated commit. It did not independently compile the repository or verify the predecessor report’s proofs. The needed predecessor patterns have therefore been reproved in the present manuscript. Literature checking was focused on surreal foundations, Kanovei–Shelah indexing, and class recursion and well-orders; it was not sufficient to certify historical originality.

B The finite checks as shipped

[merge] The programs of sources 11 and 12 use only the Python standard library and check finite instances; they verify no transfinite, class-theoretic or cardinal statement. Rerun on copies on 2 October 2026 (Python 3.14.4, Windows), each reproduced its recorded output exactly apart from line endings (both write CRLF on Windows; compare with `diff -strip-trailing-cr`): `code/11-raw-orders-finite_checks.py` prints the content of `data/11-raw-orders-finite_checks_results.txt` (465 sign-code pairs, 79,800 word-code pairs, 266,272 pairs of permutations, 867 adjacent), and `code/12-singular-finite_checks.py--output` writes the content of `data/12-singular-finite_checks.json` (180,905 cases). Always pass an output path outside the report directory. The delivered `code/12-singular-build.sh` changes into `code/` and then calls `code/finite_checks.py`; it builds nothing in this layout.

C The other sources’ title pages and abstracts

[merge] The results these summaries announce are printed in the body; the summaries are kept for the record.

Source 12

Title-page summary. Main distinction. An arbitrary class well-order of **No** need not have order type Ord. Nevertheless, two such well-orders have an *elementarily definable largest common*

labelled initial segment, possibly a proper class. This supplies a lexicographic comparison in GBC without invoking general class well-order comparability or ETR.

Principal quantitative result. Put $\mathbf{No}_{<\kappa} = \{s : \alpha \rightarrow \{-, +\} : \alpha < \kappa\}$ and $\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}$, for an infinite initial cardinal κ . The minimal-length permutation order satisfies

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases}$$

where $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$ and $\simeq_e$ means mutual order-embeddability, not isomorphism.

Status. This is an AI-assisted, unrefereed research manuscript. Proofs are supplied for the stated results; historical priority is not claimed. The accompanying exact finite tests do not verify the transfinite arguments. No new Lean formalization is claimed.

Abstract. We study the lexicographic order of well-orderings of the surreal numbers, carefully distinguishing bounded sets of surreals, set-like global well-orders, and arbitrary global class well-orders. For the full class problem, ordinal-indexed enumerations are insufficient. We construct the largest common labelled initial segment of two well-orders directly from their relations, prove that comparison of their next elements defines a linear order, and show that this construction needs only elementary class comprehension once the well-orders exist. Over GB with set choice, the existence of any class well-order of the surreals is equivalent to global choice. The collection of all these well-orders is not itself an ordinary class of GBC or Kelley–Morse theory; a diagonal theorem rules out any uniform exhaustive coding by a class of sets.

For set-like global well-orders we prove density, interpolation for set-indexed cuts, local universality for class linear orders, and mutual embeddability with the lexicographic comparison of binary classes on Ord. For unrestricted global well-orders we classify adjacent pairs: they arise precisely from finite residual permutations after a common, necessarily proper-class, prefix. At a birthday cutoff κ , we obtain an exact regular/singular dichotomy for the binary coding rank of minimal-length enumerations. We also establish the exact ordinal and binary-cube embedding spectra for the full set-sized order. Finally, a full-class model based on an inaccessible rank explains exactly how the bounded and global theories fit together, and provides a rigorous example in which the full order and the set-like order have the same cardinality and ordinal spectrum but different coding ranks. Explicit proof-assistant interfaces and further research questions are included.

D Crosswalk of the sources’ numbered statements

[merge] Each row gives a numbered statement of a source (numbered as in that source’s own PDF), and where it is printed or recorded here. A note records a statement proved here by the same argument; a section labelled “route” prints a different argument. Labels are the source’s own with its prefix, or `swo:sub:number` for unlabelled statements.

D.1 Source 11 (base)

No.	Statement in source 11	Printed here as
3.1	Definition	Definition 5.1 , p. 10
3.2	Lemma (Deleting a set)	Lemma 5.2 , p. 11
3.3	Proposition	Proposition 5.4 , p. 13

No.	Statement in source 11	Printed here as
4.1	Theorem (Cardinality and exact ordinal spectrum)	Theorem 6.1, p. 14
4.2	Proposition (Exactly where jumps occur)	Proposition 6.2, p. 15
5.1	Corollary	Corollary 7.1, p. 18
5.2	Lemma	Lemma 7.2, p. 18
6.1	Theorem (Explicit sign coding)	Theorem 9.1, p. 22
6.2	Proposition (Birthday bounds)	Proposition 9.2, p. 22
7.1	Theorem (Exact local character)	Theorem 10.1, p. 23
7.2	Corollary	Corollary 10.2, p. 23
8.1	Theorem (Word-cut classification)	Theorem 11.1, p. 24
8.2	Corollary (Successor words)	Corollary 11.2, p. 24
9.1	Proposition	Proposition 12.1, p. 25
10.1	Theorem (Existence and global choice)	Theorem 13.1, p. 26
11.1	Lemma (Unique first disagreement)	Lemma 14.1, p. 28
11.2	Theorem (Raw lexicographic order)	Theorem 14.2, p. 29
11.3	Proposition (Convex prefix cones)	Proposition 14.3, p. 29
12.1	Theorem (Canonical initial part)	Theorem 15.1, p. 32
12.2	Example (Two proper blocks)	Example 15.2, p. 32
12.3	Proposition (Convex fibers)	Proposition 15.3, p. 32
12.4	Remark (Singleton fibers and iteration)	Remark 15.4, p. 33
13.1	Lemma (Extreme enumerations)	Lemma 16.1, p. 33
13.2	Theorem (Exact adjacency criterion)	Theorem 16.2, p. 33
13.3	Example (An adjacent pair of type $\text{Ord} + 2$)	Example 16.3, p. 34
14.1	Definition	Definition 18.1, p. 37
14.2	Definition	Definition 18.2, p. 37
14.3	Lemma (Completion of an arbitrary set prefix)	Lemma 18.3, p. 38
14.4	Theorem (Prefix interpolation)	Theorem 18.5, p. 39
14.5	Corollary	Corollary 18.7, p. 40
14.6	Corollary (The class of realizations of a set cut)	Corollary 18.8, p. 40
14.7	Proposition	Proposition 18.9, p. 40
14.8	Proposition (Prefix cylinders)	Proposition 18.11, p. 41
14.9	Proposition (Approximants at every point)	Proposition 18.12, p. 41
14.10	Corollary (Set saturation is not Dedekind completeness)	Corollary 18.14, p. 42
14.11	Theorem	Theorem 18.15, p. 42
14.12	Corollary	Corollary 18.18, p. 44
14.13	Theorem (No exhaustive uniform class of codes)	Theorem 18.19, p. 44
14.14	Theorem (An omitted-range cut)	Theorem 18.22, p. 45
14.15	Example (Two omitted points produce adjacent raw orders)	Example 18.23, p. 46
14.16	Corollary (Exactly which raw set cuts admit bounded completions)	Corollary 18.24, p. 46
15.1	Theorem (Nonembedding in a full universe)	Theorem 19.1, p. 48
15.2	Theorem	Theorem 19.2, p. 49
16.1	Research question	Research question 22.1, p. 55
16.2	Research question	Research question 22.2, p. 55
16.3	Research question	Research question 22.3, p. 56
16.4	Research question	Research question 22.4, p. 56
16.5	Research question	Research question 22.5, p. 56
16.6	Research question	Research question 22.6, p. 56
16.7	Research question	Research question 22.7, p. 57
16.8	Research question	Research question 22.8, p. 57
16.9	Research question	Research question 22.9, p. 57

No.	Statement in source 11	Printed here as
16.10	Research question	Research question 22.10, p. 57

D.2 Source 12

No.	Statement in source 12	Printed here as
3.1	Definition (Common labelled prefix)	Definition 14.4, p. 30
3.2	Lemma (Maximal-prefix lemma)	Lemma 14.5, p. 30
3.3	Definition (Full lexicographic comparison)	Definition 14.6, p. 31
3.4	Theorem (Full class lexicographic order)	Theorem 14.7, p. 31
3.5	Remark (Why this does not prove class order-type comparability)	Remark 14.8, p. 31
3.6	Example (A disagreement after every ordinary ordinal)	Example 14.9, p. 31
3.7	Corollary (Agreement with enumeration lexicography)	Corollary 14.10, p. 31
4.1	Theorem (Choice equivalence)	Theorem 13.2, p. 27
4.2	Remark (The explicit set-choice hypothesis)	Remark 13.3, p. 28
5.1	Theorem (No exhaustive class-coded family)	Theorem 18.20, p. 45
5.2	Corollary (No uniform set-code solution)	Corollary 18.21, p. 45
6.1	Lemma (Extending a set prefix)	Lemma 18.4, p. 38
6.2	Lemma (Avoiding a forbidden set)	Lemma 5.3, p. 11
6.3	Theorem (Set-indexed interpolation)	Note 18.6, p. 39
6.4	Proposition (Cofinality and local character)	Note 18.13, p. 42
6.5	Lemma (Surreal universality, with the size of the recursion exposed)	Lemma 5.5, p. 13
6.6	Lemma (Deleting a set does not change the numerical class order)	Lemma 5.6, p. 13
6.7	Theorem (Local class universality and cylinders)	Note 18.10, p. 41
7.1	Lemma (A self-delimiting numerical code)	Lemma 18.16, p. 43
7.2	Theorem (Binary-class equimorphism)	Theorem 18.17, p. 43
8.1	Definition (A prefix cylinder in the full order)	Definition 16.4, p. 35
8.2	Proposition (Residual-order decomposition)	Proposition 16.5, p. 35
8.3	Theorem (Exact adjacency criterion)	Theorem 16.6, p. 35
8.4	Corollary (Proper-class prefixes are responsible)	Corollary 16.7, p. 35
9.1	Lemma (Prefix-freeness of exhaustive enumerations)	Lemma 6.3, p. 16
9.2	Lemma (Strict binary hierarchy)	Lemma 6.4, p. 16
9.3	Lemma (Characteristic-cut coding)	Lemma 6.5, p. 17
9.4	Lemma (Pair orientation coding)	Lemma 6.6, p. 17
9.5	Lemma (No μ^+ -long monotone chain)	Lemma 6.7, p. 17
9.6	Theorem (Cardinality and exact spectra)	Theorem 6.8, p. 17
9.7	Remark (What the rank does not say)	Remark 6.9, p. 18
10.1	Lemma (Row scheduler)	Lemma 8.1, p. 19
10.2	Lemma (Surreal cone decomposition)	Lemma 8.2, p. 19
10.3	Theorem (Exact minimal-slice dichotomy)	Theorem 8.3, p. 20
10.4	Corollary (A complete cube test for the minimal slice)	Corollary 8.4, p. 20
10.5	Example (An unconditional singular example)	Example 8.5, p. 21
10.6	Example (Regular cutoffs)	Example 8.6, p. 21
11.1	Theorem (Exact external identification)	Theorem 20.1, p. 51
14.1	Research question (All fixed enumeration types)	Research question 22.11, p. 58

No.	Statement in source 12	Printed here as
14.2	Research question (Noncardinal birthday bounds)	Research question 22.12 , p. 58
14.3	Research question (Isomorphism, not merely equimorphism)	Research question 22.13 , p. 58
14.4	Research question (The hierarchy of proper-class blocks)	Research question 22.14 , p. 58
14.5	Research question (The exact weak-choice strength)	Remark 22.15 , p. 58
14.6	Research question (Elementary common-prefix theory)	Research question 22.16 , p. 58
14.7	Research question (Automorphisms and finite condensation)	Research question 22.17 , p. 59
14.8	Research question (Dependence on the class realization)	Research question 22.18 , p. 59
14.9	Research question (Compatibility between birthday cutoffs)	Research question 22.19 , p. 59

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- [10] ProveIt contributors, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`; read by source 12 at revision 63bb8b1d9 and by source 11 at ae7c1e6aa. It proves minimum-birthday and simplest-separator uniqueness; source 12 did not rebuild it.
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